

ACHIEVE!



maths

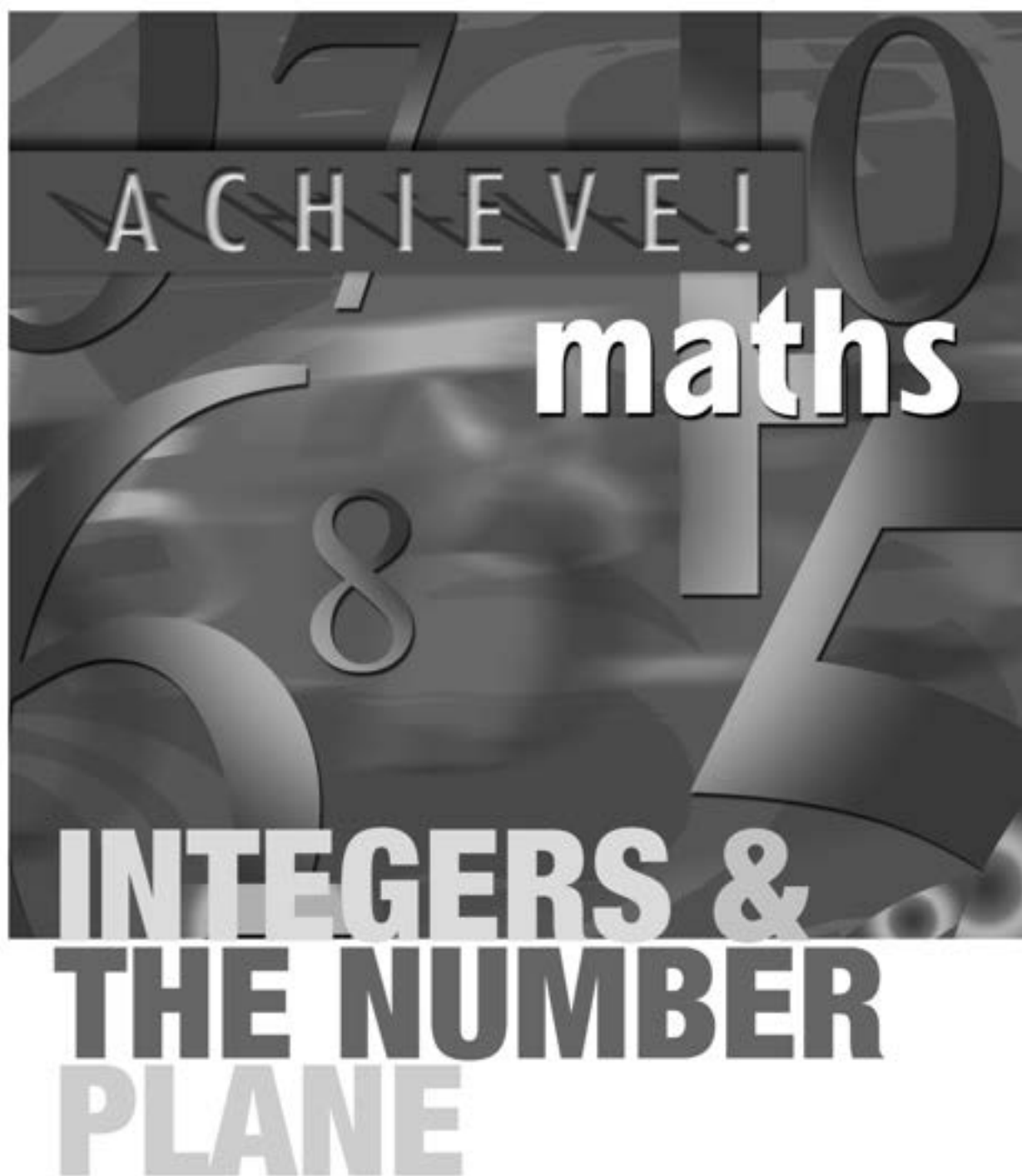
**INTEGERS &
THE NUMBER
PLANE**

Paul Steenson



Made for the
Australian Curriculum





Paul Steenson

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Introduction

The *Achieve!* series of Maths titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students. It is aimed at students who, because of poor literacy skills, struggle to read and comprehend regular classroom materials. Many of these students did not master the basics of a topic when it was first introduced. Therefore, some activity sheets in this book revisit content that would usually be covered in Year 5 or 6. Some activities are more challenging than others. The activity sheets are also suitable as supplementary material for regular classes. They may also provide useful revision in preparation for state-wide or national numeracy testing.

This book is divided into six units, each of which focuses on a topic or set of skills to do with integers or the number plane. The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other activity sheets. These documents can be easily edited to suit the individual needs of students. Note that photocopying and printing copies from the CD-ROM may change the size of diagrams. Teachers can use the material in a variety of ways. For example, a unit could be taught in sequential lessons with a whole class or with students in groups of varying sizes. Alternatively, one or more activity sheets can be used as support material, for individual homework, assessment or revision. Teachers will need to select carefully if individual activity sheets are used as stand-alone tasks, as some activities build on knowledge gained earlier in the unit. The Teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the activity sheets.

Prior knowledge – This refers to the skills and knowledge that are required for students to complete the tasks.

Background – This section provides a range of different types of useful information to help teach this unit. It can include explanations of key concepts and terminology or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Activity sheets and answers – A summary of the content of each activity sheet is provided, along with teaching tips and selected answers to assist with time-efficient marking.

Extension ideas – Additional activities or discussion ideas are suggested to recap on the main skills, knowledge or terminology taught, as well as ways to extend, assess or consolidate student understanding.

Australian Curriculum alignment

See inside front cover

Teacher's notes

OBJECTIVES

- Recognise the direction and magnitude of an integer.
- Place positive and negative numbers on a number line.
- Arrange a set of integers in ascending order.
- Interpret the use of directed numbers in a real-world context, e.g. the rise and fall of temperature.

PRIOR KNOWLEDGE

Students should be familiar with a number line using positive numbers. They should be able to compare, order and calculate with positive numbers.

BACKGROUND

There are various formats for recording directed numbers such as raised signs. The simpler format used in these activity sheets is generally less confusing and is recommended.

Note that the number -2 should be read as 'negative 2' not 'minus 2'.

STARTER ACTIVITY

A discussion of minimum and maximum temperatures around the world, using data from the daily newspaper or the internet, could be a useful activity to introduce negative numbers.

ACTIVITY SHEETS

1.1 Number skills check-up

This activity sheet gives students practice using the four basic operations with positive whole numbers. It is designed as a starter activity to this unit of work. Encourage students to complete this activity sheet without using a calculator.

Answers

1 a 5 b 9 c 9 d 8 e 7 f 12 g 11
h 13 i 11 j 11 k 12 l 15 m 12 n 11
o 14 p 13 q 14 r 17 s 17 t 16

2 a 3 b 1 c 3 d 4 e 6 f 3 g 8
h 8 i 5 j 7 k 6 l 7 m 4 n 7
o 9 p 11 q 14 r 10 s 11 t 12

3 a

+	2	4	7	5	9
1	3	5	8	6	10
3	5	7	10	8	12
0	2	4	7	5	9
6	8	10	13	11	15
8	10	12	15	13	17

b

+	3	6	5	9	10
2	5	8	7	11	12
5	8	11	10	14	15
4	7	10	9	13	14
8	11	14	13	17	18
7	10	13	12	16	17

4 a 6 b 10 c 12 d 16 e 21 f 24 g 16
h 30 i 0 j 25 k 12 l 18 m 90 n 48
o 36 p 40 q 42 r 81 s 48 t 72

5 a 4 b 5 c 4 d 6 e 4 f 7 g 8
h 5 i 4 j 8 k 7 l 6 m 8 n 9
o 9 p 6 q 9 r 8 s 7 t 9

Teacher's notes

CONTINUED

6 a

x	3	1	4	7	6
2	6	2	8	14	12
4	12	4	16	28	24
5	15	5	20	35	30
8	24	8	32	56	48
9	27	9	36	63	54

b

x	4	2	5	8	3
3	12	6	15	24	9
7	28	14	35	56	21
6	24	12	30	48	18
9	36	18	45	72	27
5	20	10	25	40	15

1.2 Number lines

This activity sheet gives students practice in using number lines with positive numbers. They are asked to list the numbers marked on a number line. They are also asked to complete a number line to show a given set of positive numbers. Time lines are included as examples.

Answers

- 1 a 1, 2, 3, 5, 7, 11 b 2, 4, 6, 8, 10
c 4, 5, 6, 7, 8 d 5, 7, 9, 11
3 a 12 b 15 c 6 d 3 e 15
4 a 9 am b 9 pm c 7 am d 4 pm

1.3 Negative numbers

In this activity sheet, temperatures below zero are used to introduce negative numbers.

Answers

- 1 a 35°C b 15°C c 0°C d 2°C
e -5°C f -1°C
2 a true b true c true d false
e true f false
3 a 30°C b 25°C c 5°C d -2°C
4 a Mount Buller b 3°C

1.4 Positive and negative numbers

In this activity sheet, students compare numbers using a number line. They are asked to identify which number is larger or smaller. They are also asked to complete a number line to show a given set of integers and list them in ascending order.

Answers

- 1 a 5 b 7 c 4 d 0 e -2 f -3
g 5 h -3
2 a 3 b 4 c -4 d -6 e -3 f -5
g -7 h -4
3 a 3 b 4 c 3 d 4 e 4 f 5
g 5 h 6 i 8 j 8 k 12 l 11
4 a true b false c true d true e true
f false g true h false i true j true
k true l true
5 a -2, 1, 3, 5 b -4, -2, 1, 5
c -4, -3, 3, 4 d -4, -1, 2, 3
e -5, -3, -1, 0 f -6, -3, 0, 2
g -4, -2, -1, 3, 5 h -6, -3, 2, 4, 5

EXTENSION IDEAS

Teachers could remind students that addition and subtraction are inverse operations, as are multiplication and division. Encourage students to use this information as an aid to mental computation.

1.1 Number skills check-up

The numbers we use for counting are the **whole numbers**.

They are written using the digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

The four basic operations that can be carried out with numbers are:

- **Addition** (+ means *add*, *plus*, find the *sum* or *total*)
- **Subtraction** (– means *minus*, *subtract* or *take away*)
- **Multiplication** (× means *times*, *multiply* or find the *product*)
- **Division** (÷ means *divide* or *share*)



1 Find the following sums.

- | | | | |
|--------------------------------------|--------------------------------------|---------------------------------------|---------------------------------------|
| a $3 + 2 = \underline{\quad}$ | b $6 + 3 = \underline{\quad}$ | c $2 + 7 = \underline{\quad}$ | d $3 + 5 = \underline{\quad}$ |
| e $4 + 3 = \underline{\quad}$ | f $5 + 7 = \underline{\quad}$ | g $8 + 3 = \underline{\quad}$ | h $7 + 6 = \underline{\quad}$ |
| i $6 + 5 = \underline{\quad}$ | j $4 + 7 = \underline{\quad}$ | k $3 + 9 = \underline{\quad}$ | l $7 + 8 = \underline{\quad}$ |
| m $8 + 4 = \underline{\quad}$ | n $5 + 6 = \underline{\quad}$ | o $9 + 5 = \underline{\quad}$ | p $4 + 9 = \underline{\quad}$ |
| q $6 + 8 = \underline{\quad}$ | r $8 + 9 = \underline{\quad}$ | s $11 + 6 = \underline{\quad}$ | t $12 + 4 = \underline{\quad}$ |

2 Complete these subtractions.

- | | | | |
|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| a $9 - 6 = \underline{\quad}$ | b $5 - 4 = \underline{\quad}$ | c $8 - 5 = \underline{\quad}$ | d $7 - 3 = \underline{\quad}$ |
| e $8 - 2 = \underline{\quad}$ | f $10 - 7 = \underline{\quad}$ | g $11 - 3 = \underline{\quad}$ | h $12 - 4 = \underline{\quad}$ |
| i $11 - 6 = \underline{\quad}$ | j $12 - 5 = \underline{\quad}$ | k $13 - 7 = \underline{\quad}$ | l $13 - 6 = \underline{\quad}$ |
| m $13 - 9 = \underline{\quad}$ | n $14 - 7 = \underline{\quad}$ | o $17 - 8 = \underline{\quad}$ | p $16 - 5 = \underline{\quad}$ |
| q $19 - 5 = \underline{\quad}$ | r $16 - 6 = \underline{\quad}$ | s $19 - 8 = \underline{\quad}$ | t $18 - 6 = \underline{\quad}$ |

3 Complete these addition squares.

a

+	2	4	7	5	9
1					
3					
0					
6					
8					

b

+	3	6		9	
2					12
		11			
4					
8			13		
	10				

1.1 Number skills check-up

CONTINUED

4 Complete these multiplications.

a $3 \times 2 = \underline{\quad}$ **b** $2 \times 5 = \underline{\quad}$ **c** $6 \times 2 = \underline{\quad}$ **d** $4 \times 4 = \underline{\quad}$

e $7 \times 3 = \underline{\quad}$ **f** $6 \times 4 = \underline{\quad}$ **g** $2 \times 8 = \underline{\quad}$ **h** $3 \times 10 = \underline{\quad}$

i $6 \times 0 = \underline{\quad}$ **j** $5 \times 5 = \underline{\quad}$ **k** $4 \times 3 = \underline{\quad}$ **l** $9 \times 2 = \underline{\quad}$

m $10 \times 9 = \underline{\quad}$ **n** $8 \times 6 = \underline{\quad}$ **o** $4 \times 9 = \underline{\quad}$ **p** $8 \times 5 = \underline{\quad}$

q $7 \times 6 = \underline{\quad}$ **r** $9 \times 9 = \underline{\quad}$ **s** $12 \times 4 = \underline{\quad}$ **t** $9 \times 8 = \underline{\quad}$

5 Complete these divisions. Remember, $5 \overline{)40}$ means $40 \div 5$.

a $12 \div 3 = \underline{\quad}$ **b** $10 \div 2 = \underline{\quad}$

c $16 \div 4 = \underline{\quad}$ **d** $18 \div 3 = \underline{\quad}$

e $20 \div 5 = \underline{\quad}$ **f** $28 \div 4 = \underline{\quad}$

g $24 \div 3 = \underline{\quad}$ **h** $30 \div 6 = \underline{\quad}$

i $24 \div 6 = \underline{\quad}$ **j** $32 \div 4 = \underline{\quad}$

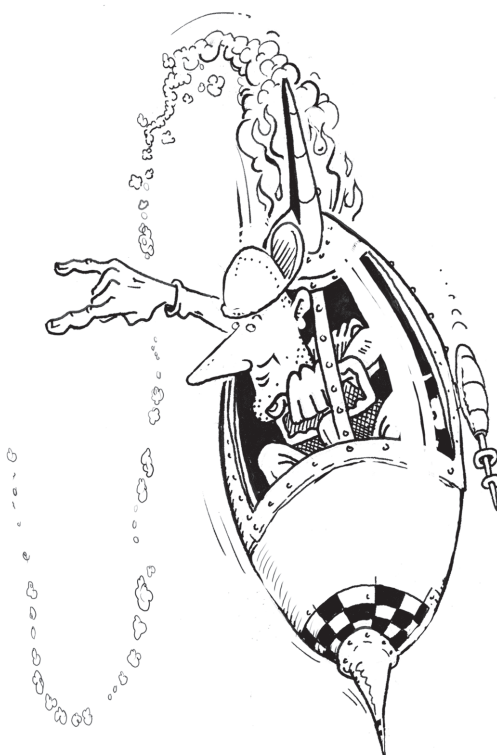
k $35 \div 5 = \underline{\quad}$ **l** $42 \div 7 = \underline{\quad}$

m $5 \overline{)40} = \underline{\quad}$ **n** $4 \overline{)36} = \underline{\quad}$

o $3 \overline{)27} = \underline{\quad}$ **p** $6 \overline{)36} = \underline{\quad}$

q $7 \overline{)63} = \underline{\quad}$ **r** $6 \overline{)48} = \underline{\quad}$

s $8 \overline{)56} = \underline{\quad}$ **t** $9 \overline{)81} = \underline{\quad}$



6 Complete these multiplication squares.

a

$\times$	3	1	4	7	6
2					
4					
5					
8					
9					

b

$\times$	4	2	5	8	
3					9
		14			
6					
9					
	20				

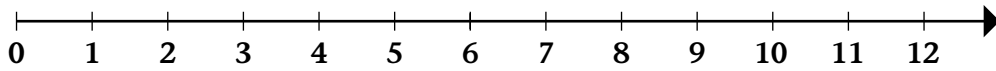
1.2 Number lines

The counting numbers 1, 2, 3, 4, 5, 6,..... are called the **positive integers**.

They can be arranged in order along a **number line**.

Zero is placed at the left hand end. The positive integers extend to the right.

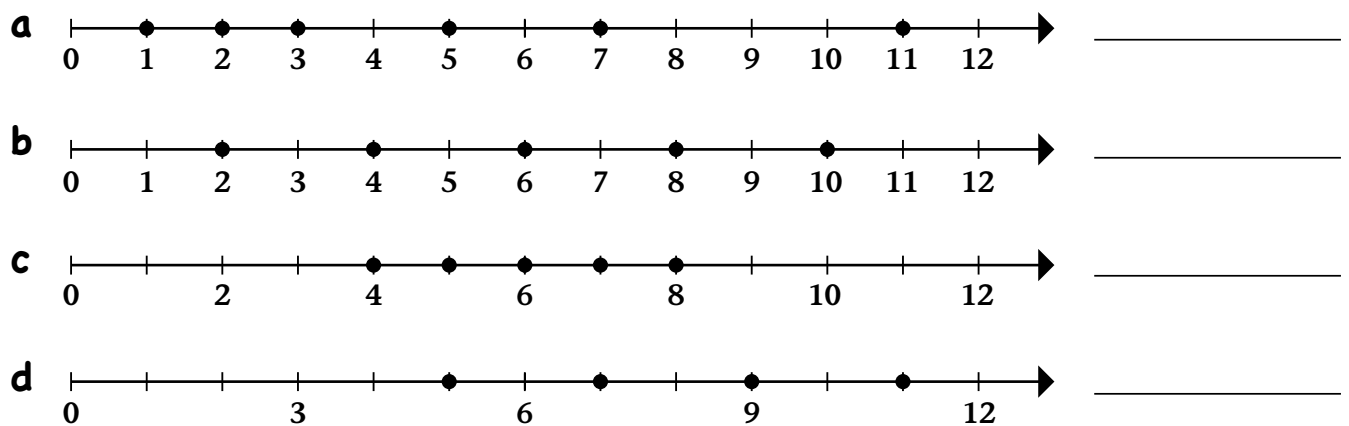
They are arranged in **ascending order**.



A number line can be used to compare the position and size of numbers.

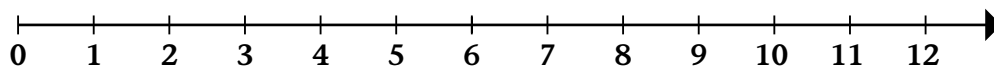
1 List the numbers marked by the shaded circles on these number lines.

Note that not every position on a number line needs to be labelled.

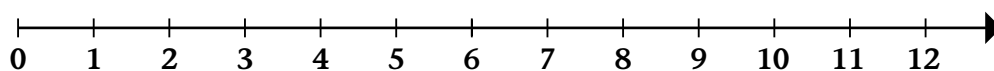


2 Draw shaded circles to show the given numbers on these number lines.

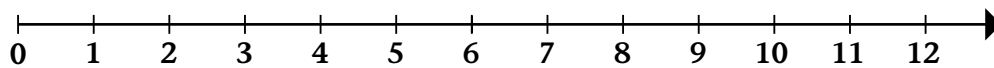
a The numbers 2, 5, 8 and 11



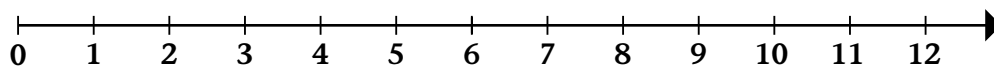
b The **even** numbers between 3 and 9



c The integers (whole numbers) between 2 and 7



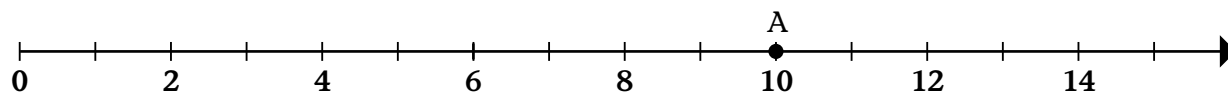
d The positive integers less than 6



1.2 Number lines

CONTINUED

- 3 Gemma drew this number line.
She marked the position of a number with the letter A.

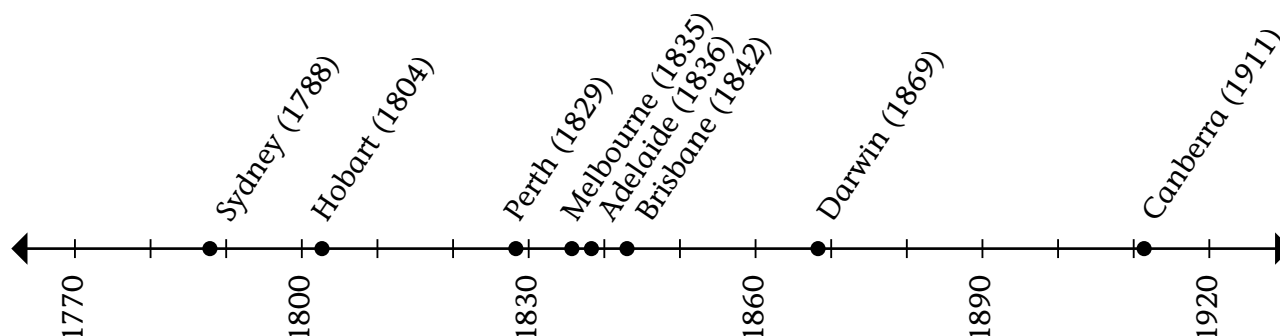


Use the starting position shown in the diagram to answer each question.

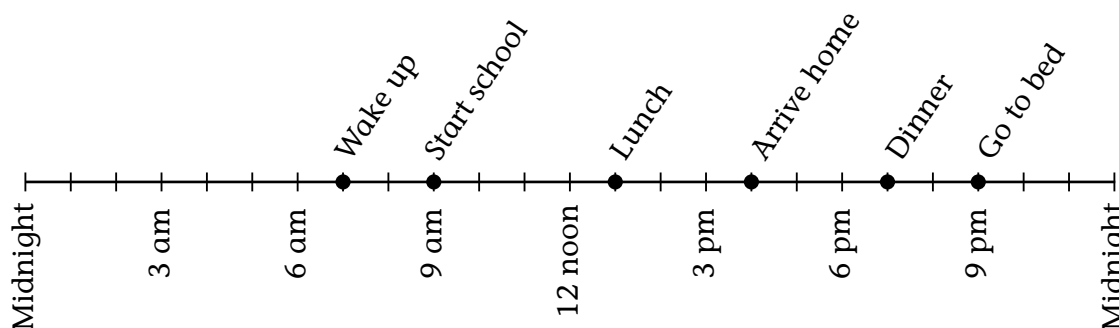
Where will A be if it is moved:

- a 2 places to the right? _____ b 5 places to the right? _____
c 4 places to the left? _____ d 7 places to the left? _____
e 3 places to the left and then 8 places to the right? _____

Time lines are examples of number lines. This time line shows the years when the various capital cities in Australia were founded. It starts at 1770, the year Captain Cook sailed along the east coast. Each unit on the scale represents 10 years.



- 4 This time line shows a full day from midnight to midnight.
Each unit on the scale represents 1 hour. Peter marked his normal routine on it.



At what time does Peter:

- a start school? _____ b go to bed at night? _____
c wake up in the morning? _____ d arrive home? _____

1.3 Negative numbers

In some parts of Australia, the temperature can be more than 40°C on very hot days.

In winter, some places experience temperatures below zero.

Temperatures below zero are written with a minus or negative sign ($-$), for example -5°C .

1 Circle the **hotter** or **higher** temperature in each pair.

- | | |
|---|--|
| a 35°C 20°C | b 10°C 15°C |
| c 0°C -10°C | d 2°C -5°C |
| e -5°C -10°C | f -3°C -1°C |

2 In these questions, $>$ means 'is greater than' or 'is hotter than' and $<$ means 'is less than' or 'is colder than'.

Answer true or false.

- a** $25^{\circ}\text{C} > 18^{\circ}\text{C}$ _____
- b** $-5^{\circ}\text{C} < 0^{\circ}\text{C}$ _____
- c** $10^{\circ}\text{C} > -10^{\circ}\text{C}$ _____
- d** $5^{\circ}\text{C} < -15^{\circ}\text{C}$ _____
- e** $-3^{\circ}\text{C} > -6^{\circ}\text{C}$ _____
- f** $-8^{\circ}\text{C} < -12^{\circ}\text{C}$ _____

3 The temperature in Canberra is 10°C .

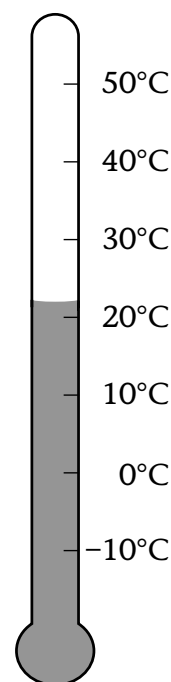
What is the temperature in:

- a** Darwin, if it is 20°C hotter than Canberra? _____
- b** Rockhampton, if it is 15°C hotter than Canberra? _____
- c** Hobart, if it is 5°C colder than Canberra? _____
- d** Thredbo, if it is 12°C colder than Canberra? _____

4 The temperature at Perisher Valley was -2°C last night.

At Mount Buller it was -5°C .

- a** Which town was colder? _____
- b** What is the difference between these temperatures? _____



1.4 Positive and negative integers

Positive numbers are greater than zero.

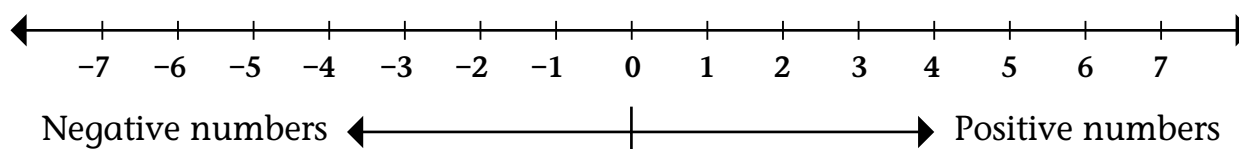
They extend to the **right** of zero on a number line.

Negative numbers are less than zero.

They extend to the **left** of zero on a number line.

The number zero is neither positive nor negative.

All the positive and negative whole numbers and zero are called **integers**.



1 Circle the **larger number** in each pair.

The larger number is further to the **right** on a number line.

- | | | | |
|--------------------|--------------------|-------------------|--------------------|
| a 3 and 5 | b 7 and 2 | c 4 and -3 | d -7 and 0 |
| e -6 and -2 | f -3 and -4 | g 5 and -6 | h -7 and -3 |

2 Circle the **smaller number** in each pair.

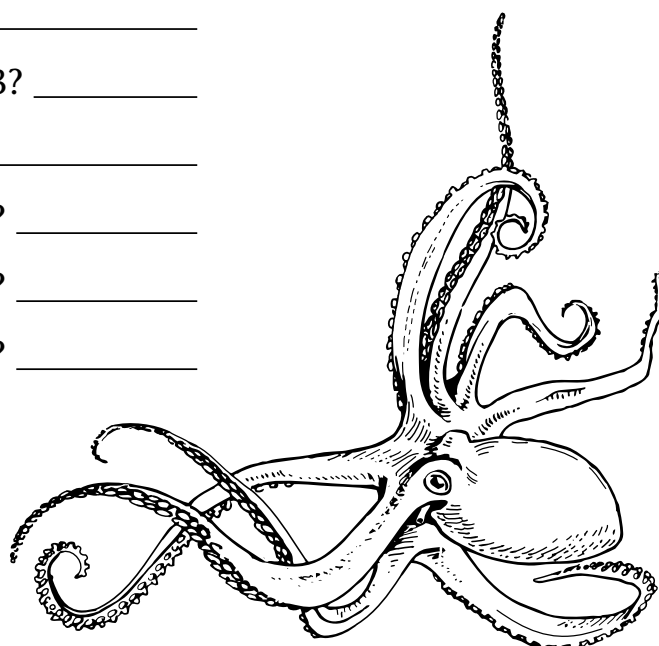
The smaller number is further to the **left** on a number line.

- | | | | |
|--------------------|-------------------|-------------------|--------------------|
| a 7 and 3 | b 4 and 5 | c 0 and -4 | d -6 and -1 |
| e -3 and -2 | f -5 and 2 | g 3 and -7 | h -4 and 4 |

3 We say the distance between 0 and 1 on a number line is 1 unit.

How many units apart are:

- | | |
|---------------------------|---------------------------|
| a 2 and 5? _____ | b 3 and 7? _____ |
| c -5 and -2? _____ | d -7 and -3? _____ |
| e -4 and 0? _____ | f 0 and 5? _____ |
| g -3 and 2? _____ | h 4 and -2? _____ |
| i -5 and 3? _____ | j -4 and 4? _____ |
| k 6 and -6? _____ | l 7 and -4? _____ |



1.4 Positive and negative integers

CONTINUED

An inequality sign ($<$ or $>$) always points towards the smaller number.

EXAMPLES

$6 > 2$ 6 is **greater** than 2

$-3 < -1$ -3 is **less** than -1

4 Are these statements true or false?

a $3 > 1$ _____

b $7 > 8$ _____

c $5 < 9$ _____

d $5 > -2$ _____

e $-1 < 0$ _____

f $7 < -9$ _____

g $-6 < -4$ _____

h $-3 > 6$ _____

i $-6 < 5$ _____

j $4 > -7$ _____

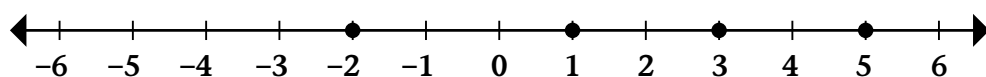
k $-6 < -2$ _____

l $-5 > -10$ _____

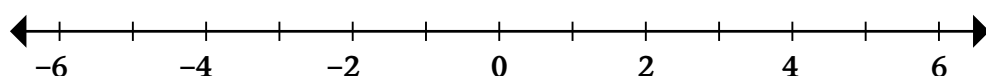
5 Plot each set of integers on the number line provided.
Then, write the numbers in order from smallest to largest.
Part a has been done for you.

a 1, 3, -2, 5

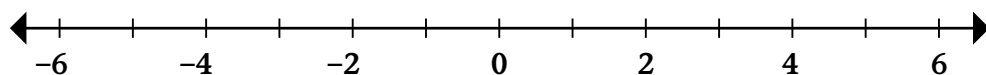
-2, 1, 3, 5



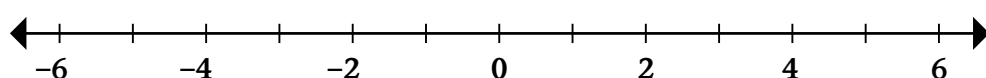
b -2, 1, 5, -4



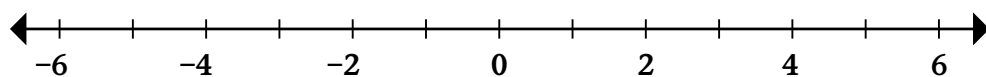
c -3, 3, 4, -4



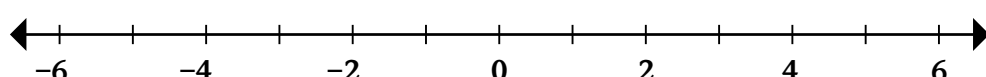
d -4, 2, -1, 3



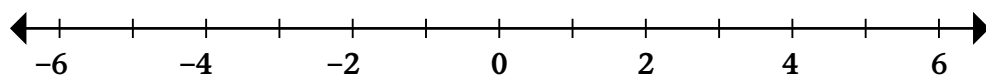
e -1, -3, -5, 0



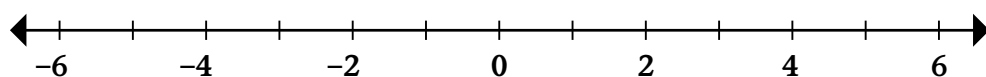
f -6, 0, -3, 2



g 5, -2, -4, 3, -1



h -6, 4, 5, -3, 2



Teacher's notes

OBJECTIVES

- Add and subtract positive and negative integers.
- Multiply and divide positive and negative integers.
- Interpret the different meanings (direction or operation) for the + or – signs depending on the context.
- Simplify an expression involving grouping symbols and apply the order of operations rules.
- Solve word problems involving positive and negative numbers

PRIOR KNOWLEDGE

Students should be able to perform the basic operations with positive whole numbers using mental arithmetic or a calculator. They should recognise addition and subtraction as inverse operations.

BACKGROUND

There are various formats for recording directed numbers such as raised signs. The simpler format used in these activity sheets is generally less confusing and is recommended.

STARTER ACTIVITY

A short quiz using the four basic operations with positive whole numbers, or choosing the larger or smaller of two numbers, could be a useful starter activity. Alternatively, the class could be organised in two teams and play a game of *Maths Jeopardy*, using questions involving the four basic operations with positive numbers. If an incorrect answer is given, the point value is subtracted from the team's score.

ACTIVITY SHEETS

2.1 Adding integers

In this activity sheet, students add integers using a number line. To add a positive number they count to the right (the positive direction). To add a negative number they add to the left (the negative direction). They are also given word problems which involve positive and negative numbers.

Answers

- 1 a 7 b 4 c -1 d 3 e -3 f -3 g 3
h -5 i -8 j -7 k 5 l -5 m -11 n 1
o -11 p 5 q -7 r -2 s -9 t -10 u 0
v 2 w -1 x -11
- 2 8°C 3 3 km 4 -\$115
- 5 a 1, 1, -2, -2, 1, 1, 1, -2, 1, 1 b 1
- 6 6 runs 7 -5 8 35 m above 9 Down \$56

2.2 Subtracting integers

In this activity sheet, students subtract integers using a number line. They are also given word problems which involve positive and negative numbers.

Answers

- 1 a 4 b 3 c 5 d -5 e -3 f -5 g -7
h -8 i -6 j -2 k -7 l -11 m 7 n 7
o 4 p -2 q 12 r 6 s 3 t -12 u -3
v 0 w -3 x 4
- 2 15 3 -9°C 4 17°C 5 7 hours
- 6 a 7 hours b 12 hours 7 2 316 m 8 9 237 m

2.3 Multiplying integers

In this activity sheet, students apply the rules for multiplying positive and negative numbers.

Encourage them to learn these rules and complete the questions without using a calculator.

Answers

- 1 a 21 b 20 c -6 d -15 e -12 f -28 g 30
h 24 i 45 j 42 k -40 l -24 m -56 n 36
o 36 p 35 q 0 r -56 s -81 t 48 u -27
v -32 w -72 x 54

Teacher's notes

CONTINUED

2.4 Division with integers

In this activity sheet, students apply the rules for division with positive and negative numbers.

Encourage them to learn these rules and complete the questions without using a calculator.

Answers

- 1 a 8 b 6 c -3 d -8 e -6 f -7 g -4
h 4 i 3 j 9 k -9 l -7 m -6 n 6
o -5 p -4 q -5 r 5 s -8 t -7 u 9
v -6 w 8 x -8
- 2 a -6 b -7 c -4 d -8 e -9 f -8 g 6
h 6 i 7

2.5 Mixed calculations

In this activity sheet, students complete various calculations involving positive and negative numbers, including calculations involving grouping symbols and more than one operation.

Answers

- 1 a 1 b -6 c 2 d 2 e 4 f -7
g 0 h 2 i -2
- 2 a -2 b -8 c 4 d 6 e -3 f 8
g -10 h -17 i -12
- 3 a -6 b -12 c -15 d -35 e 20 f 28
g -45 h 32 i 0
- 4 a -5 b -5 c -4 d -1 e 5 f 4
g -8 h -6 i 3
- 5 a -4 b 1 c 24 d 15 e 5 f -2
g -11 h -6 i -14 j -30 k -9 l -5
m 7 n 6 o 32 p -77 q -12 r 9
- 6 a 6 b 2 c -6 d 0 e -8 f 0
g 2 h -2 i -6 j -18 k 12 l -2
m 4 n 16 o -10 p 3 q 18 r 0

2.6 Further problems involving integers

This activity sheet presents further word problems involving positive and negative numbers.

Answers

- 1 -7°C 2 88 3 -5°C 4 34°C
- 5 Debit \$622 (or $-\$622$)
- 6 a 95 m b 315 m c 220 m

EXTENSION IDEAS

Students could be given practice using a calculator to perform mixed calculations with integers.

The class could play a game of *Guess the Number*. The teacher could ask questions involving one operation or more. For example: *What number, when I add 6, gives 4? What number, when I add -5 then subtract 2, gives 12?* This could be done as a class quiz or with students organised in teams.

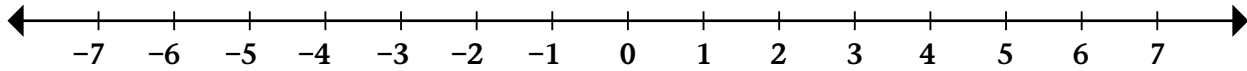
2.1 Adding integers

The temperature last night was -3 degrees.

This morning it has increased by 5 degrees. What is the temperature?

To calculate the answer we need to add -3 and 5.

To add positive and negative numbers, refer to a number line.

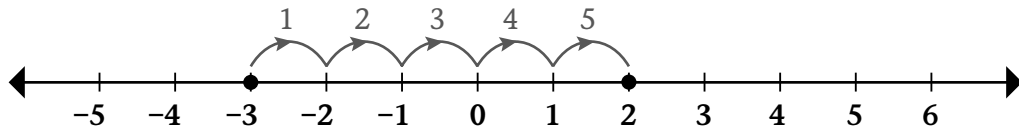


To add a positive number, move to the right.

To add a negative number, move to the left.

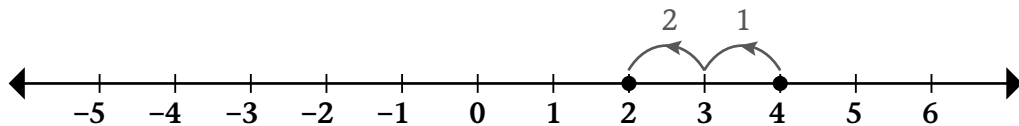
EXAMPLES

$-3 + 5$ To add -3 and 5, start at -3 on the number line.
Count 5 places to the right (the positive direction).



The answer is 2, so $-3 + 5 = 2$

$4 + (-2)$ To add 4 and -2 , start at 4 on the number line.
Count 2 places to the left (the negative direction).



The answer is 2, so $4 + (-2) = 2$

Note that $4 + (-2)$ is the same as $4 - 2$.

1 Complete these additions.

You can use the number line at the top of this page to help.

a $3 + 4 =$ _____ **b** $-3 + 7 =$ _____ **c** $-4 + 3 =$ _____

d $6 + (-3) =$ _____ **e** $2 + (-5) =$ _____ **f** $4 + (-7) =$ _____

g $-2 + 5 =$ _____ **h** $-1 + (-4) =$ _____ **i** $-2 + (-6) =$ _____

j $-4 + (-3) =$ _____ **k** $-3 + 8 =$ _____ **l** $0 + (-5) =$ _____

m $-7 + (-4) =$ _____ **n** $7 + (-6) =$ _____ **o** $-8 + (-3) =$ _____

p $-5 + 10 =$ _____ **q** $5 + (-12) =$ _____ **r** $7 + (-9) =$ _____

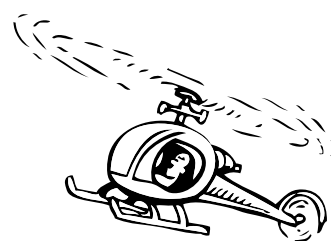
s $-9 + 0 =$ _____ **t** $-5 + (-5) =$ _____ **u** $8 + (-8) =$ _____

v $-7 + 9 =$ _____ **w** $8 + (-9) =$ _____ **x** $-4 + (-7) =$ _____

2.1 Adding integers

CONTINUED

- 2** The overnight temperature was -2°C .
By midday it had risen 10°C .
What was the temperature at midday? _____
- 3** Emily walked 6 kilometres east.
She then walked 9 kilometres west.
How far is Emily from her starting point? _____
- 4** Joe's bank statement showed a balance of $-\$355$.
He owed the bank $\$355$. He made a deposit of $\$240$.
What is the new balance? _____
- 5** In a quiz, contestants score 1 point for a correct answer. They lose 2 points for each wrong answer. Kurt's answers for ten questions were:
correct, correct, wrong, wrong, correct, correct, correct, wrong, correct, correct.
- a** Write Kurt's score for each question. _____
- b** What was Kurt's total score for the quiz? _____
- 6** On five consecutive balls in indoor cricket, Bryan scored:
2 runs, 1 run, 6 runs, run out (lose 5 runs), 2 runs.
What was his total score from these five balls? _____
- 7** Positive and negative numbers are often used to keep score in a game of golf. Kate had these scores on nine holes:
par (0), bogey (+1), birdie (-1), par (0), bogey (+1), eagle (-2),
par (0), birdie (-1), albatross (-3).
What is her total score? _____
- 8** A helicopter is flying over a valley.
It rises 30 metres to fly over some trees.
It then falls 50 metres, rises 25 metres,
falls 10 metres and rises 40 metres.
How far is the helicopter above or below its original position? _____
- 9** Michael has a savings account.
He made the following transactions during the past month:
deposit $\$236$, withdraw $\$415$, deposit $\$165$, deposit $\$284$, withdraw $\$326$.
How had his balance changed at the end of the month? _____

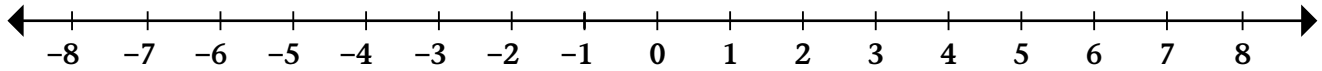


2.2 Subtracting integers

The temperature was 12°C at 1 pm. By midnight it had dropped to -3°C .
What is the difference between these temperatures?

To calculate a difference we subtract: $12 - (-3)$.

To subtract positive and negative numbers, refer to a number line.



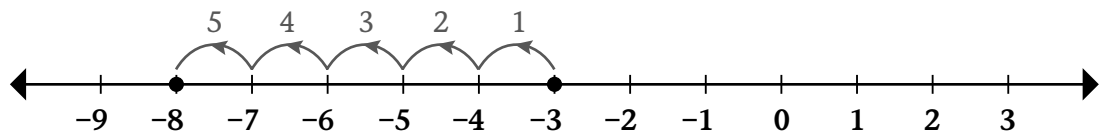
To subtract a positive number, move to the left.

To subtract a negative number, do the opposite: move to the **right**.

We can **replace** $--$ with $+$.

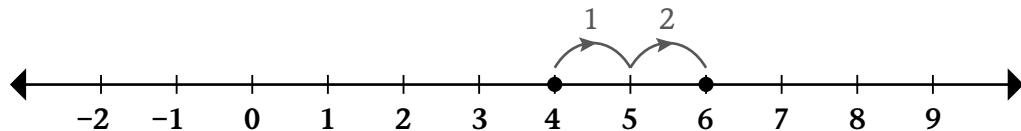
EXAMPLES

$-3 - 5$ To subtract 5 from -3 , start at -3 on the number line.
Count 5 places to the **left** (in the negative direction).



The answer is -8 , so $-3 - 5 = -8$

$4 - (-2)$ To subtract -2 from 4, start at 4 on the number line.
Count 2 places to the **right** (the opposite of the negative direction).



The answer is 6, so $4 - (-2) = 6$

1 Complete these subtractions.

You can use the number line at the top of this page to help.

- | | | |
|------------------------------|-----------------------------|-------------------------------|
| a $6 - 2 =$ _____ | b $10 - 7 =$ _____ | c $8 - 3 =$ _____ |
| d $4 - 9 =$ _____ | e $3 - 6 =$ _____ | f $0 - 5 =$ _____ |
| g $-5 - 2 =$ _____ | h $-4 - 4 =$ _____ | i $-6 - 0 =$ _____ |
| j $4 - 6 =$ _____ | k $2 - 9 =$ _____ | l $-4 - 7 =$ _____ |
| m $5 - (-2) =$ _____ | n $4 - (-3) =$ _____ | o $-1 - (-5) =$ _____ |
| p $-6 - (-4) =$ _____ | q $7 - (-5) =$ _____ | r $-3 - (-9) =$ _____ |
| s $-5 - (-8) =$ _____ | t $-8 - 4 =$ _____ | u $-8 - (-5) =$ _____ |
| v $-9 - (-9) =$ _____ | w $9 - 12 =$ _____ | x $-7 - (-11) =$ _____ |

2.2 Subtracting integers

CONTINUED

- 2 The temperature on a day in winter reached a maximum of 12°C . It dropped down to -3°C overnight.

How many degrees did the temperature drop? _____



- 3 The daytime temperature in Stockholm was -2°C . The next night, the temperature dropped 7 degrees. What was the temperature that night? _____
- 4 On a winter's day in Moscow, the maximum temperature was -4°C . The lowest temperature overnight was -21°C . What is the difference between these temperatures? _____
- 5 The time in Auckland (New Zealand) is 2 hours ahead of the time in Melbourne. The time in Bangkok (Thailand) is 3 hours behind Melbourne time. What is the time difference between Auckland and Bangkok? _____

- 6 The table compares the local time in these cities with Australian Eastern Standard Time (AEST). AEST is the time in the eastern states of Australia.

City	Hours difference from AEST
Bangkok	-3
London	-10
New York	-15

- a What is the time difference between London and Bangkok? _____
- b What is the time difference between Bangkok and New York? _____

- 7 The summit of Mount Kosciuszko is the highest point in Australia. It is 2 230 metres above sea level. Death Valley in California, USA, is 86 metres **below** sea level. What is the difference in altitude between these two points? (**Altitude** refers to the height above or below sea level.) _____
- 8 The highest point on Earth is Mount Everest (8 840 metres above sea level). The lowest point on land is the Dead Sea in the Jordan Valley (397 metres **below** sea level). What is the difference in altitude between these two points? _____

2.3 Multiplying integers

John made a video of his sister reversing her car out of the driveway and then driving forward along the street.

He later played the video **backwards**. The car appeared to reverse along the street and drive **forwards** into the driveway.

Film direction	The car was driven	The car appears to travel
Forward (+)	Forward (+)	Forward (+)
Forward (+)	Reverse (-)	Reverse (-)
Backwards (-)	Forward (+)	Reverse (-)
Backwards (-)	Reverse (-)	Forward (+)



The rules for multiplying positive and negative numbers are:

- positive $\times$ positive = positive $(+) \times (+) = +$
- positive $\times$ negative = negative $(+) \times (-) = -$
- negative $\times$ positive = negative $(-) \times (+) = -$
- negative $\times$ negative = positive $(-) \times (-) = +$

Two like signs give a positive answer.

Two unlike (different) signs give a negative answer.

EXAMPLES

$$2 \times 5 = 10$$

$$4 \times (-3) = -12$$

$$-3 \times 5 = -15$$

$$-4 \times (-5) = 20$$

1 Complete these multiplications.

a $3 \times 7 =$ _____

b $4 \times 5 =$ _____

c $3 \times (-2) =$ _____

d $5 \times (-3) =$ _____

e $-6 \times 2 =$ _____

f $-4 \times 7 =$ _____

g $6 \times 5 =$ _____

h $-4 \times (-6) =$ _____

i $-5 \times (-9) =$ _____

j $-6 \times (-7) =$ _____

k $-5 \times 8 =$ _____

l $3 \times (-8) =$ _____

m $-8 \times 7 =$ _____

n $-6 \times (-6) =$ _____

o $9 \times 4 =$ _____

p $-7 \times (-5) =$ _____

q $-4 \times 0 =$ _____

r $8 \times (-7) =$ _____

s $9 \times (-9) =$ _____

t $-8 \times (-6) =$ _____

u $-3 \times 9 =$ _____

v $8 \times (-4) =$ _____

w $-9 \times 8 =$ _____

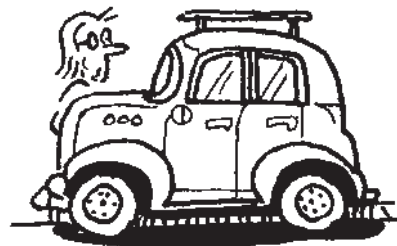
x $-9 \times (-6) =$ _____

2.4 Division with integers

Division is the **reverse operation** to **multiplication**.

When you divide with positive and negative numbers, the rules are:

- positive $\div$ positive = positive $(+) \div (+) = +$
- positive $\div$ negative = negative $(+) \div (-) = -$
- negative $\div$ positive = negative $(-) \div (+) = -$
- negative $\div$ negative = positive $(-) \div (-) = +$



Two like signs give a positive answer.

Two unlike (different) signs give a negative answer.

These are the same as the rules we have for multiplication.

EXAMPLES

$$10 \div 2 = 5$$

$$12 \div (-3) = -4$$

$$-15 \div 5 = -3$$

$$-20 \div (-4) = 5$$

1 Complete these divisions.

a $16 \div 2 =$ _____

b $18 \div 3 =$ _____

c $12 \div (-4) =$ _____

d $16 \div (-2) =$ _____

e $-18 \div 3 =$ _____

f $-28 \div 4 =$ _____

g $-20 \div 5 =$ _____

h $-20 \div (-5) =$ _____

i $-15 \div (-5) =$ _____

j $-27 \div (-3) =$ _____

k $36 \div -4 =$ _____

l $42 \div (-6) =$ _____

m $-24 \div 4 =$ _____

n $-30 \div (-5) =$ _____

o $25 \div (-5) =$ _____

p $32 \div (-8) =$ _____

q $-35 \div 7 =$ _____

r $-40 \div (-8) =$ _____

s $-48 \div 6 =$ _____

t $56 \div (-8) =$ _____

u $-45 \div (-5) =$ _____

v $-54 \div 9 =$ _____

w $-72 \div (-9) =$ _____

x $64 \div (-8) =$ _____

A division can be written as a fraction.

For example: $\frac{18}{3}$ means $18 \div 3$ so $\frac{18}{3} = 18 \div 3 = 6$.

2 Complete these divisions.

a $\frac{-12}{2} = -12 \div 2$

b $\frac{-21}{3} =$ _____

c $\frac{-24}{6} =$ _____

$=$ _____

$=$ _____

$=$ _____

d $\frac{32}{-4} =$ _____

e $\frac{27}{-3} =$ _____

f $\frac{40}{-5} =$ _____

$=$ _____

$=$ _____

$=$ _____

g $\frac{-48}{-8} =$ _____

h $\frac{-42}{-7} =$ _____

i $\frac{-63}{-9} =$ _____

$=$ _____

$=$ _____

$=$ _____

2.5 Mixed calculations with integers

1 Complete these additions. You can use a number line to help.

- To add a positive number, move to the right.
- To add a negative number, move to the left.

a $-3 + 4 =$ _____	b $-1 + (-5) =$ _____	c $4 + (-2) =$ _____
d $5 + (-3) =$ _____	e $-2 + 6 =$ _____	f $-3 + (-4) =$ _____
g $7 + (-7) =$ _____	h $-6 + 8 =$ _____	i $-8 + 6 =$ _____

2 Complete these subtractions.

Remember, to subtract a negative number, replace $- (-)$ with $+$.

a $3 - 5 =$ _____	b $-3 - 5 =$ _____	c $8 - 4 =$ _____
d $4 - (-2) =$ _____	e $-6 - (-3) =$ _____	f $5 - (-3) =$ _____
g $-7 - 3 =$ _____	h $-8 - 9 =$ _____	i $-6 - 6 =$ _____

3 Complete these multiplications. When you multiply two integers:

- two like signs give a positive answer
- two unlike (different) signs give a negative answer.

a $3 \times (-2) =$ _____	b $4 \times (-3) =$ _____	c $-3 \times 5 =$ _____
d $-5 \times 7 =$ _____	e $-4 \times (-5) =$ _____	f $-7 \times -4 =$ _____
g $-9 \times 5 =$ _____	h $-8 \times (-4) =$ _____	i $-5 \times 0 =$ _____

4 Complete these divisions. When you divide with integers:

- two like signs give a positive answer
- two unlike (different) signs give a negative answer.

a $-10 \div 2 =$ _____	b $-15 \div 3 =$ _____	c $12 \div (-3) =$ _____
d $6 \div (-6) =$ _____	e $-20 \div (-4) =$ _____	f $-24 \div (-6) =$ _____
g $-32 \div 4 =$ _____	h $48 \div (-8) =$ _____	i $-27 \div (-9) =$ _____

5 Complete these calculations.

a $5 - 9 =$ _____	b $-6 + 7 =$ _____	c $6 \times 4 =$ _____
d $-5 \times (-3) =$ _____	e $8 + (-3) =$ _____	f $-7 - (-5) =$ _____
g $-4 + (-7) =$ _____	h $-12 \div 2 =$ _____	i $-6 - 8 =$ _____
j $5 \times (-6) =$ _____	k $36 \div (-4) =$ _____	l $-8 + 3 =$ _____
m $-56 \div (-8) =$ _____	n $3 - (-3) =$ _____	o $-8 \times (-4) =$ _____
p $-7 \times 11 =$ _____	q $-8 - 4 =$ _____	r $-81 \div (-9) =$ _____

2.5 Mixed calculations with integers

CONTINUED

When a calculation involves **more than one operation**:

- **Step 1:** Simplify inside any grouping symbols (brackets).
- **Step 2:** Work any $\times$ or $\div$ from left to right.
- **Step 3:** Work any $+$ or $-$ from left to right.

EXAMPLES

The first step is shown in **bold**.

$$\mathbf{3} - 8 + 2$$

$$= -5 + 2$$

$$= -3$$

$$\mathbf{-7} + 3 \times 4$$

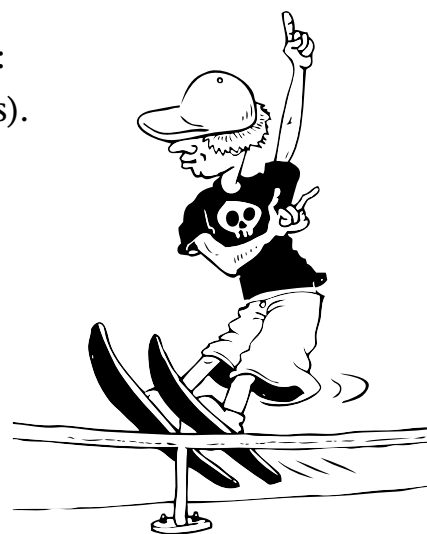
$$= -7 + 12$$

$$= 5$$

$$\mathbf{(6 - 9)} \div 3$$

$$= -3 \div 3$$

$$= -1$$



6 Complete these calculations.

a $7 - 3 + 2$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

b $6 - (-3) - 7$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

c $-4 + 7 - 9$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

d $-9 - (-6) + 3$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

e $7 - 5 \times 3$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

f $6 + 3 \times (-2)$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

g $-8 + 5 \times 2$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

h $9 - (4 + 7)$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

i $(5 - 8) \times 2$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

j $(-4 - 2) \times 3$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

k $-4 \times (3 - 6)$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

l $(4 - 12) \div 4$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

m $(-9 - 7) \div (-4)$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

n $-8 \times 2 \times (-1)$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

o $6 \times (-5) \div 3$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

p $6 \times (-4) \div (-8)$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

q $-27 \div (-3) \times 2$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

r $-4 \times (-3) - 6 \times 2$

$$= \underline{\hspace{2cm}}$$

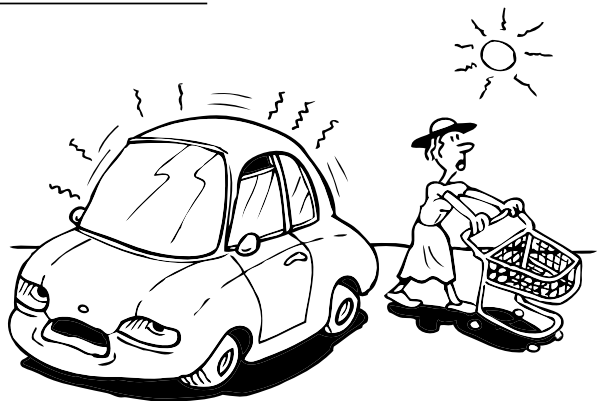
$$= \underline{\hspace{2cm}}$$

2.6 Further problems involving integers

- 1 At midday the temperature at Perisher Valley was 5°C .
The temperature dropped 2°C every hour.
What was the temperature at 6 pm? _____
- 2 In a game of indoor cricket, a team loses 5 runs off their score each time they lose a wicket. Sean's team scored 123 runs, but lost 7 wickets.
What is their final score? _____
3. The temperature was -2°C at 6 am yesterday morning.
It rose by 7°C during the morning, and dropped 10°C during the afternoon.
What was the temperature at 6 pm? _____

- 4 The average temperature in January at Alice Springs was 29°C .
At Beijing (China), the average temperature during January was -5°C .

What is the difference between these temperatures?



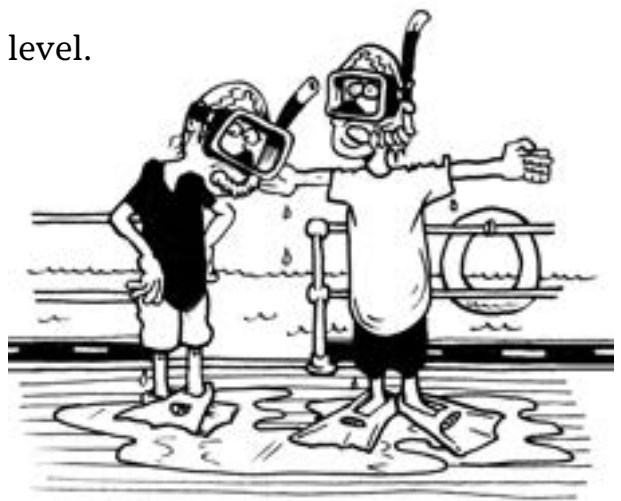
- 5 Brendan had a debit balance of \$463 in his credit card account at the beginning of the month. He owed \$463. He made a payment of \$260. Brendan then made purchases costing \$47, \$163, \$85 and \$124.
What is the new balance on his account? _____

- 6 Anita is scuba diving 70 metres below sea level.
The water is 165 metres deep.
A helicopter flying directly above Anita reports it is 150 metres above sea level.

- a How far is Anita above the ocean floor?

- b How far is the helicopter above the ocean floor?

- c What is the vertical distance between Anita and the helicopter? _____



Teacher's notes

OBJECTIVES

- Use simple maps and grids to represent position.
- Use co-ordinates to describe positions on maps and plans
- Find a place on a map or plan, given its co-ordinates.

PRIOR KNOWLEDGE

Students should be able to use everyday language to describe position and give simple directions.

BACKGROUND

Grids are used in many contexts to identify position. Simple maps and grids are used here to establish the notion of locating position using a pair of co-ordinates. While alternate grid systems may be used in students' early experience, the standard square grid system should be established prior to introducing the Cartesian number plane (named after Renee Descartes).

STARTER ACTIVITY

The teacher could ask students to describe the location of an object in the classroom. For example, 'The book is on the second shelf and fourth from the left'.

ACTIVITY SHEETS

3.1 A shopping centre

This activity sheet presents a simple plan of a shopping centre set out on grid squares. Students are asked to locate the position of various shops using a pair of co-ordinates (one letter and one number).

Answers

- 1 Chemist 2 Bank 3 E 1 4 Hairdresser
 5 a Dress shop b Deli c Newsagent d Café
 e Hot bread shop f Supermarket
 6 a A 3 b A 5 c D 4 d E 7 e D 3
 7 D 2

3.2 A map of Australia

In this activity sheet, students are given a simple map of Australia with a square grid. The positions of various cities and towns are identified using a pair of co-ordinates to locate grid squares.

Answers

- 1 Alice Springs
 2 a Esperance b Port Hedland c Grafton
 d Brisbane e Townsville f Cobar
 g Longreach h Sale i Ceduna
 j Oodnadatta
 3 J 1
 4 a J 3 b H 4 c I 8 d B 5 e C 9
 f J 10 g L 4 h J 2 i I 3 j A 7

3.3 A classroom plan

In this activity sheet, students use an ordered pair of numbers to identify positions on a classroom seating plan.

Answers

- 1 Toulou
 2 a Kara b Graeme c Claire d Kerrie
 e Quyen f Jane g Ahmed h Vince
 i Hoan j Mia
 3 (1, 4)
 4 a (1, 1) b (3, 1) c (1, 3) d (5, 1)
 e (4, 2) f (4, 4) g (6, 4) h (6, 1)
 i (3, 3) j (6, 3)
 5 No
 6 (5, 2)
 7 24
 8 Stephen

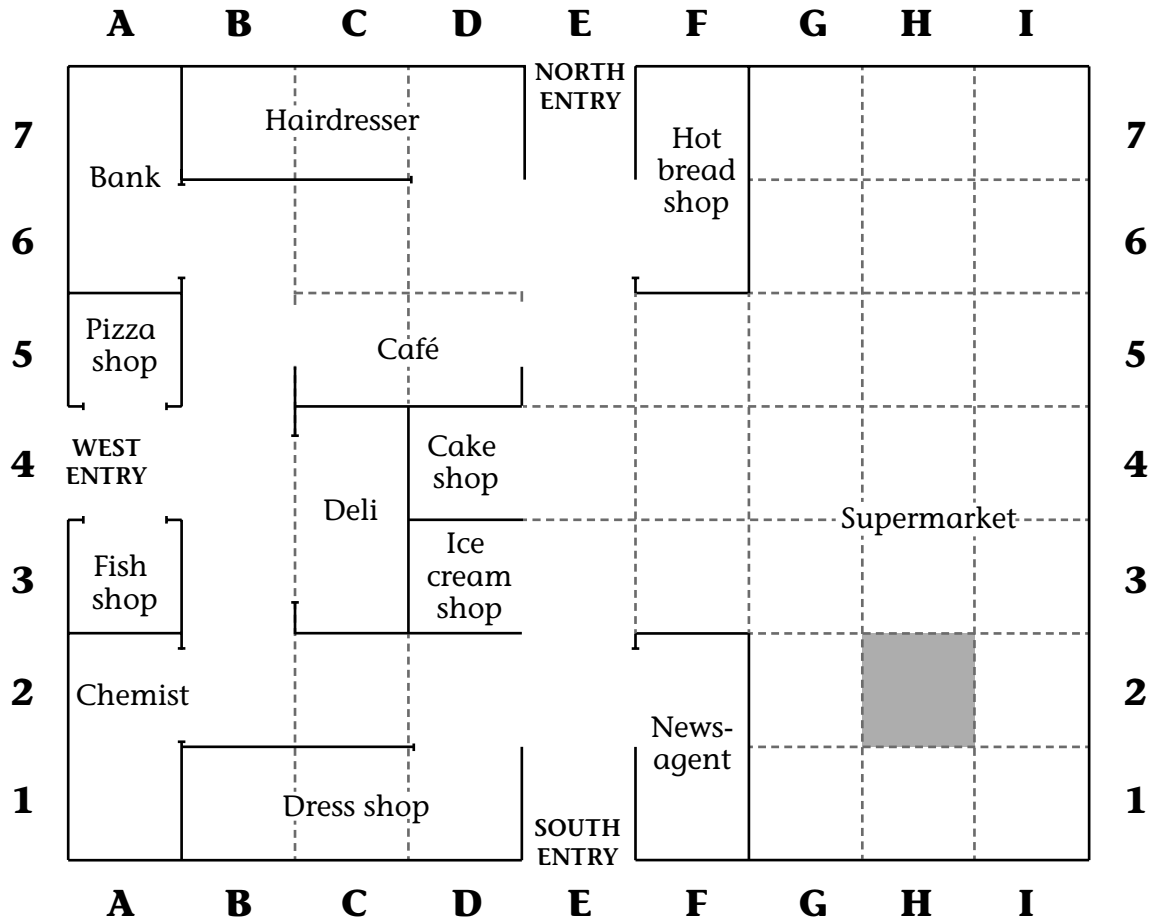
EXTENSION IDEAS

Students could be given a simple map of their school or a local shopping centre and, using a grid placed over the map, locate various positions. Students could create their own simple map and, by using a grid over the map, describe locations. Students could draw up a seating plan of their classroom and identify the position of each student using an ordered pair of numbers. They could record the ordered pairs on an alphabetical list of students.

3.1 A shopping centre

Zara and her family live near the Woodport Plaza shopping centre.

There is a plan of the centre at each entrance. A copy of this plan is shown below.



The plan uses grid squares to help us find our way around the shopping centre.

The grid squares are identified using a letter and a number.

They are called the grid **co-ordinates** of the square.

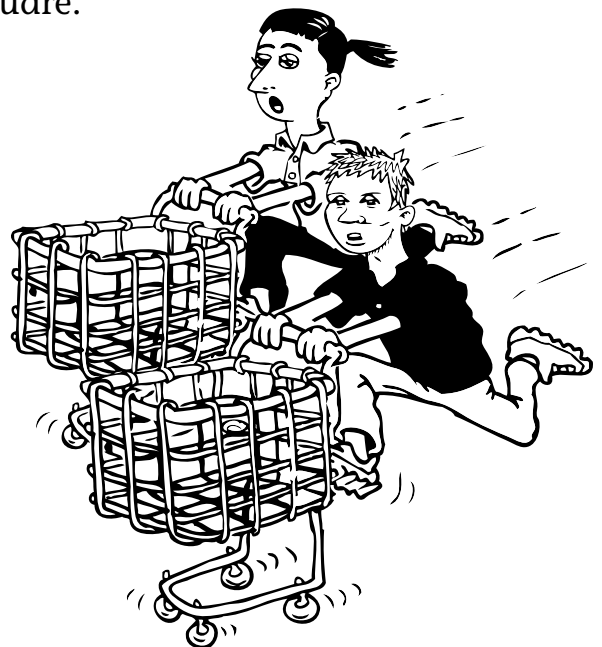
The square **H 2** has been shaded on the grid.

It is located in the supermarket.

If you enter the shopping centre

at the **west entry**, you are at **A 4** on the grid.

The fish shop has grid co-ordinates **A 3**.



3.1 A shopping centre

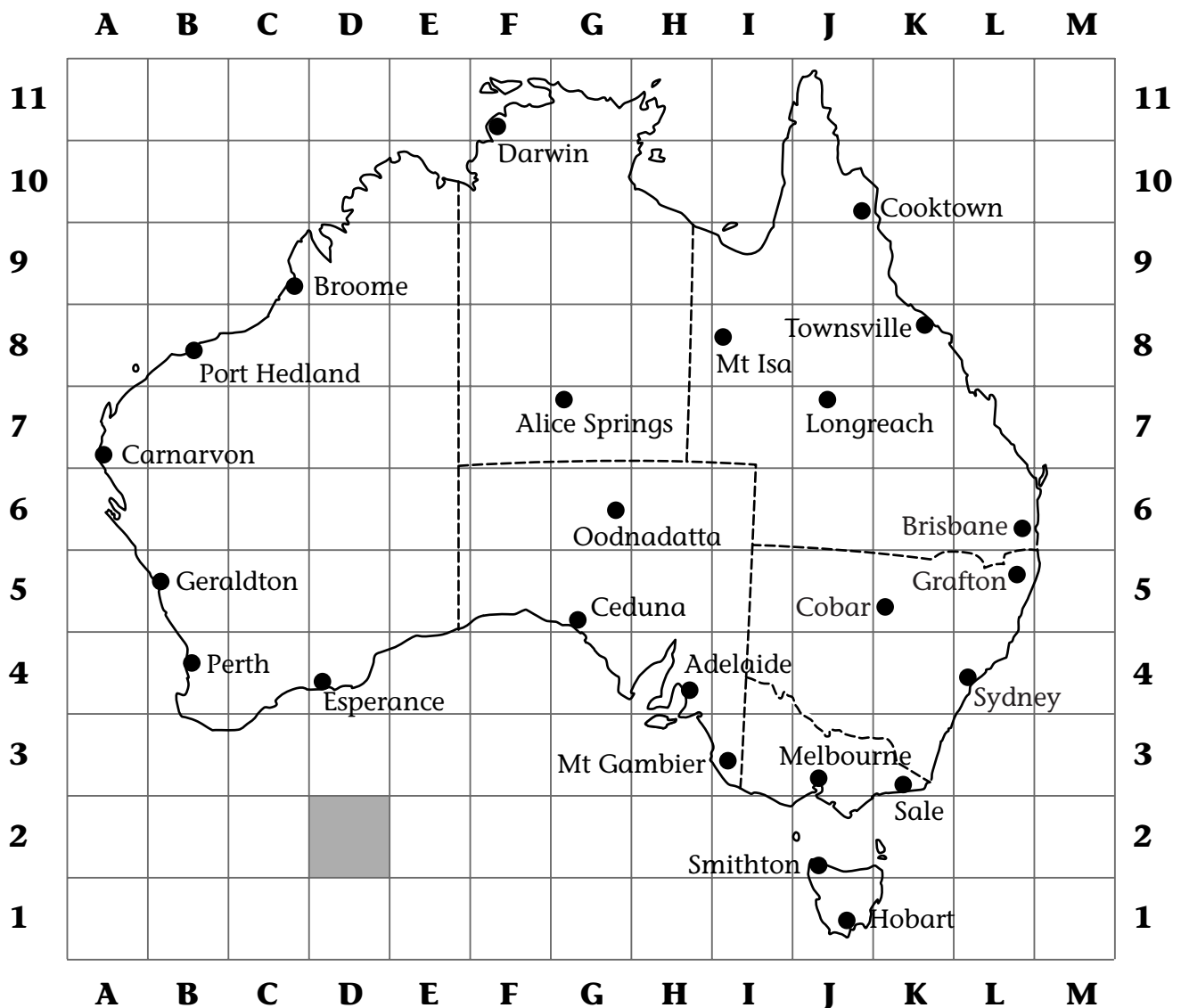
CONTINUED

- 1 Which shop is located at A 2? _____
- 2 What will you find at A 6? _____
- 3 Zara enters the shopping centre at the south entry.
In which grid square is she standing? _____
- 4 Zara wants to go to the shop at grid square D 7.
Which shop is at D 7? _____
- 5 What will you find at these co-ordinates?
 - a D 1 _____
 - b C 3 _____
 - c F 2 _____
 - d D 5 _____
 - e F 6 _____
 - f H 4 _____
- 6 What are the grid co-ordinates of:
 - a the fish shop? _____
 - b the pizza shop? _____
 - c the cake shop? _____
 - d the north entry? _____
 - e the ice-cream shop? _____
- 7 Zara needs to meet her mother outside
the entrance to the dress shop.
What are the grid co-ordinates
of the meeting place?



3.2 A map of Australia

Jared found this map of Australia in a book.



The grid squares help you to locate cities and towns.

The grid squares are identified using a letter and a number.

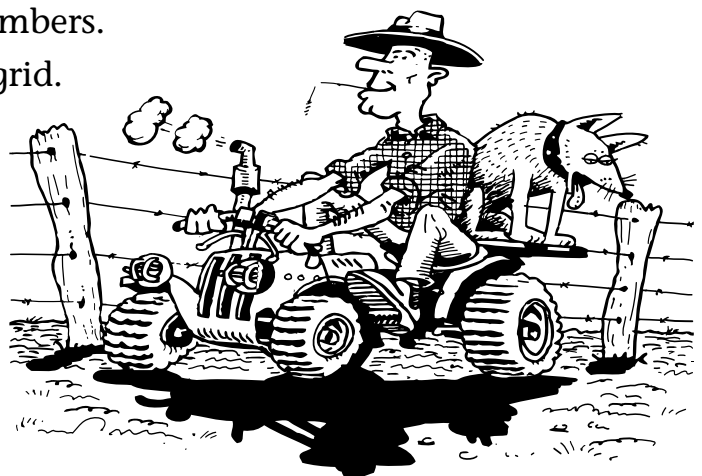
The vertical columns are labelled with letters.

The horizontal rows are labelled with numbers.

The square D 2 has been shaded on the grid.

Perth is found in square B 4.

B and 4 are called the
grid co-ordinates of this square.



3.2 A map of Australia

CONTINUED

1 Which town is located in square G 7? _____

2 Which cities or towns are found in these squares?

a D 4 _____

b B 8 _____

c L 5 _____

d L 6 _____

e K 8 _____

f K 5 _____

g J 7 _____

h K 3 _____

i G 5 _____

j G 6 _____



3 What are the grid co-ordinates for Hobart? _____

4 What are the grid co-ordinates for these cities and towns?

a Melbourne _____

b Adelaide _____

c Mt Isa _____

d Geraldton _____

e Broome _____

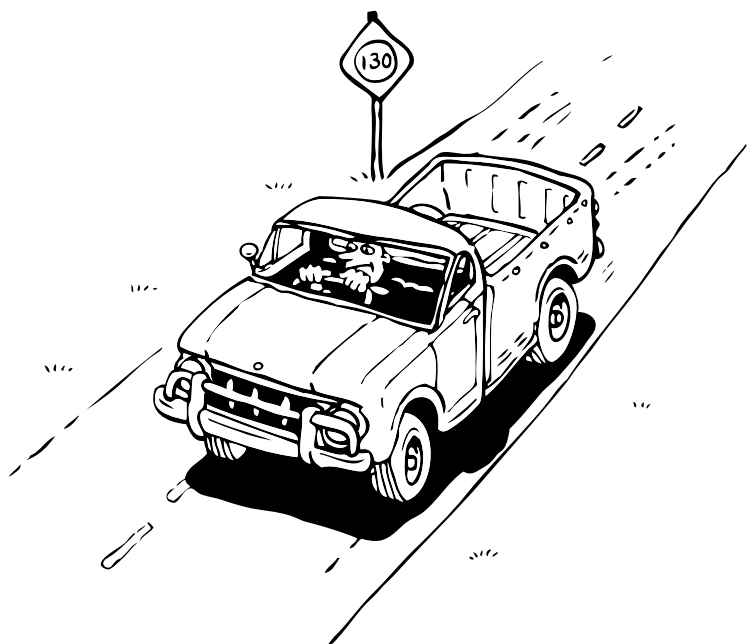
f Cooktown _____

g Sydney _____

h Smithton _____

i Mt Gambier _____

j Carnarvon _____



3.3 A classroom plan

Ms Mathie prepared a seating plan of her classroom. She uses it to learn the names of her students. It shows the desk of each student in her class.

There are 4 rows of desks. The desks are arranged in 6 blocks.

Rows	4	Brendan	Ahmed	Kerrie	Anne	Hoan	Stephen
	3	Michelle	Graeme	Hwa	Vince	Jane	Mark
	2	Kara	Quyen	Simon	Jacob		Mia
	1	Andrew	Toula	Tania	Claire	Sean	Therese
		1	2	3	4	5	6
		Blocks					

On this seating plan, the blocks are numbered 1 to 6 (from left to right).

Block 1 is closest to the door.

The rows are numbered 1 to 4.

Row 1 is at the front of the room.

Row 4 is at the back.

The location of each student can be given by an **ordered pair** of numbers.

The ordered pair can be called the **co-ordinates** of the location.

Simon's desk is located in block 3, row 2.

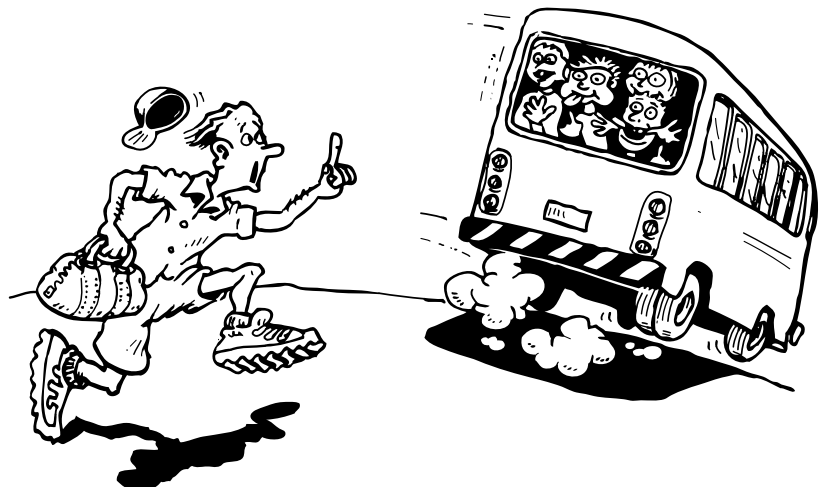
We can write this as the ordered pair (3, 2).

Always give the block number first.

Then write the row number.

Jane's desk is located at (5, 3)

– block 5, row 3.



3.3 A classroom plan

CONTINUED

1 Which student is sitting at desk (2, 1) – block 2 row 1? _____

2 Write the names of the students sitting at the following desks.

a (1, 2) _____

b (2, 3) _____

c (4, 1) _____

d (3, 4) _____

e (2, 2) _____

f (5, 3) _____

g (2, 4) _____

h (4, 3) _____

i (5, 4) _____

j (6, 2) _____



3 Brendan is sitting at the desk in block 1 row 4.

Write the ordered pair for Brendan's desk. _____

4 Write the ordered pairs for the desks of these students.

a Andrew _____

b Tania _____

c Michelle _____

d Sean _____

e Jacob _____

f Anne _____

g Stephen _____

h Therese _____

i Hwa _____

j Mark _____

5 Is there a desk position (3, 5)? _____

6 What is the ordered pair for the spare desk? _____

7 How many desks are in the classroom? _____

8 Andrew's desk is closest to the door.

Which student sits furthest from the door? _____



Teacher's notes

OBJECTIVES

- Use a square grid and an ordered pair of numbers to indicate position in a plane.
- Interpret the number plane formed from the intersection of a horizontal x -axis and a vertical y -axis.
- Identify the point of intersection of the two axes as the origin, having co-ordinates $(0, 0)$
- Read, plot and name ordered pairs on a number plane (using positive co-ordinates only).

PRIOR KNOWLEDGE

Students should be able to locate points on a grid using an ordered pair of co-ordinates.

BACKGROUND

In this unit, the notion of locating position established in Unit 3 is further developed using an ordered pair of positive numbers.

STARTER ACTIVITY

Students could be asked to follow directions similar to those in examples 1, 2 and 3 on Activity sheet 4.1. Give students a sheet of grid paper with the axes drawn for the first quadrant only. The teacher reads the instructions; for example: *Start at the bottom left hand corner. Go across 3, then up 4. Mark this point.* The answer can be displayed on an interactive whiteboard or overhead projector.

ACTIVITY SHEETS

4.1 Using co-ordinates

This activity sheet introduces the first quadrant of the number plane, using positive co-ordinates only. The terms *axes*, *ordered pair* and *co-ordinates* are introduced. Students are asked to match points and their co-ordinates.

Answers

- 1 A (1, 3) B (0, 5) C (2, 6) D (3, 3)
E (4, 0) F (5, 2) G (4, 5) H (6, 4)
- 2 a D b F c C d E e H f I g B
h G i K j A

4.2 Reading co-ordinates

In this activity sheet, students are asked to read the co-ordinates of points marked on a number plane. Encourage students to write the co-ordinates as an ordered pair of numbers using parentheses.

Answers

- 1 A (1, 3) B (5, 2) C (7, 1) D (9, 4)
E (8, 5) F (6, 4) G (2, 5) H (3, 0)
- 2 a (1, 0), (1, 1), (1, 2), (1, 3), (2, 3), (3, 3),
(4, 3), (4, 2), (4, 1), (4, 0), (3, 0), (2, 0)
b a square
- 3 a A (1, 2) B (3, 4) C (7, 2) D (3, 0)
b (3, 2)
- 4 a (5, 5) b (2, 1) c (4, 8) d (6, 2)
e (8, 8) f (11, 7) g (4, 6) h (12, 2)

4.3 Locating points

In this activity sheet, students receive practice plotting points in the first quadrant of a number plane.

Answers

- 1 a rectangle
2 an octagon
4 Meals on Wheels

EXTENSION IDEAS

Students could draw a geometric shape with the vertices on grid points. Each student, in turn, gives instructions for the other students to copy the shape. For example: *start at (0, 1), join to the point (2, 3), then (5, 4) then (4, 2) and back to (0, 1).*

4.1 Using co-ordinates

A number plane is a grid using two number lines, called **axes**.

One number line is drawn horizontal, across the page. It is called the **x-axis**.

The second number line is drawn vertically, at right angles to the x-axis. The vertical axis is called the **y-axis**.

Any position on the grid can be named using an **ordered pair** of numbers. These numbers are called the **co-ordinates** of the point.

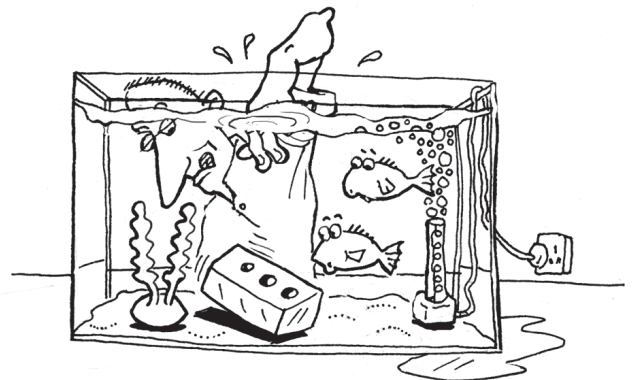
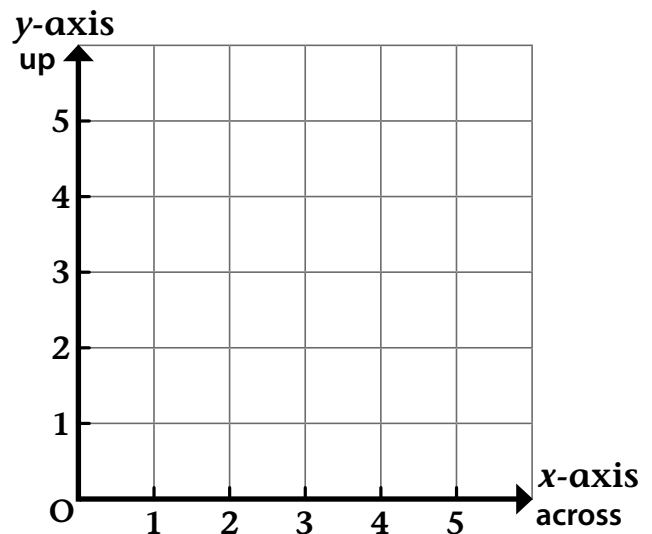
The two axes meet at the origin.

It is the point (0, 0). It is always labelled O.

To locate a point, start at the origin.

The first number tells how far to move across, along the x-axis.

The second number then tells you how far to move up.



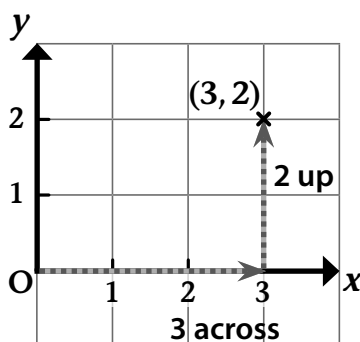
EXAMPLES

(3, 2) means 3 across, 2 up.

Start from the origin, O.

Move 3 units across.

Now, move 2 units up.



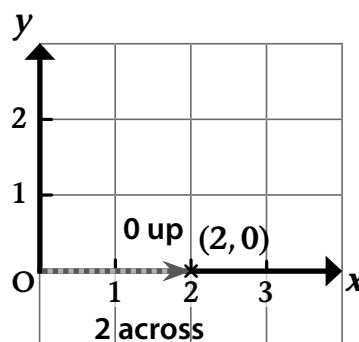
(2, 0) means 2 across, 0 up.

Start from the origin, O.

Move 2 units across.

Now, move 0 units up.

This point is located on the x-axis.



(0, 3) means 0 across, 3 up.

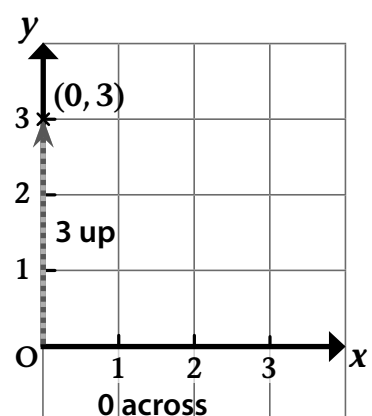
Start from the origin, O.

Move 0 units across.

(Stay at O.)

Now, move 3 units up.

This point is located on the y-axis.

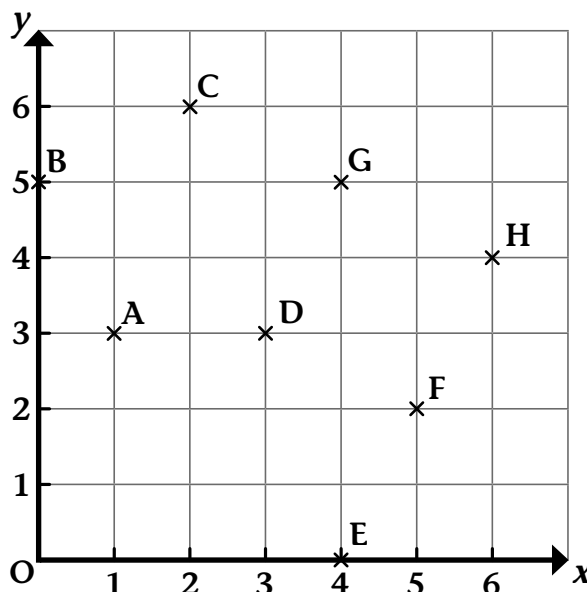


4.1 Using co-ordinates

CONTINUED

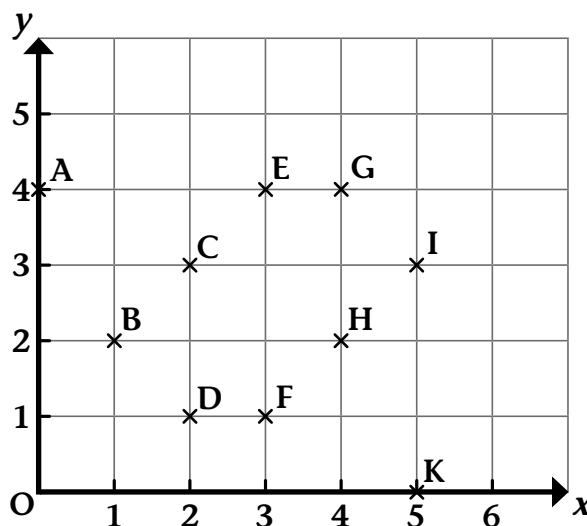
- 1 Points A to H have been labelled on this number plane.
Write the co-ordinates of each point.

A _____
 B _____
 C _____
 D _____
 E _____
 F _____
 G _____
 H _____



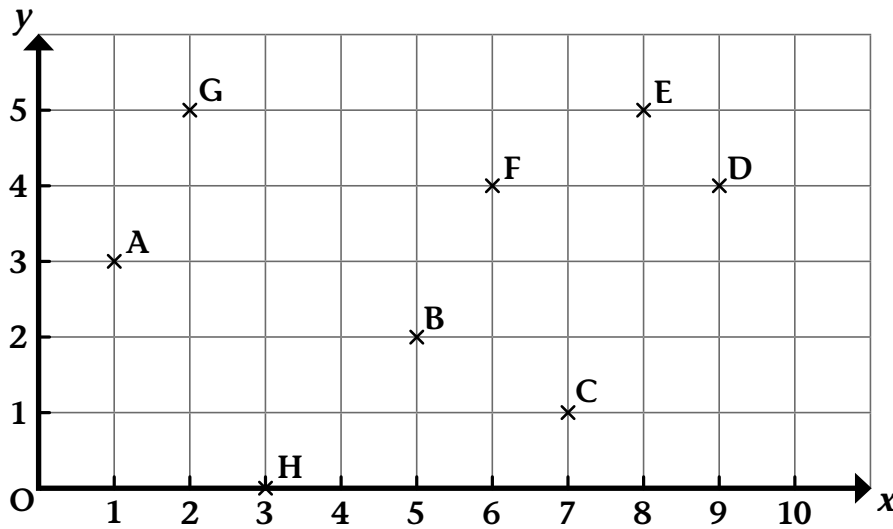
- 2 Name the points located by the following ordered pairs.

a (2, 1) _____
 b (3, 1) _____
 c (2, 3) _____
 d (3, 4) _____
 e (4, 2) _____
 f (5, 3) _____
 g (1, 2) _____
 h (4, 4) _____
 i (5, 0) _____
 j (0, 4) _____



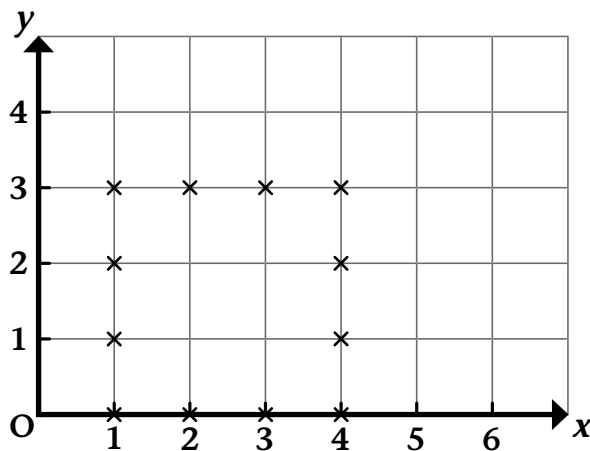
4.2 Reading co-ordinates

- 1 Write the co-ordinates of each point, A to H.



A _____ B _____ C _____
 D _____ E _____ F _____
 G _____ H _____

- 2 a List the co-ordinates of the points marked on this number plane.



- b Use a ruler to join each point to the two points closest to it.

What plane shape is formed? _____



4.2 Reading co-ordinates

CONTINUED

3 Rachel drew a kite on this grid.

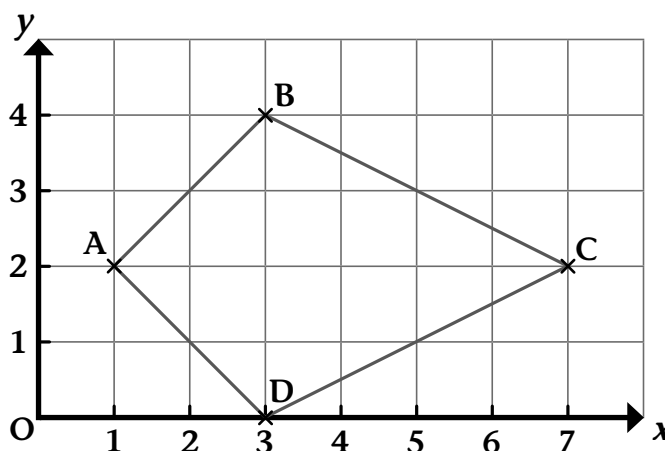
a List the co-ordinates of the vertices of the kite.

A _____

B _____

C _____

D _____

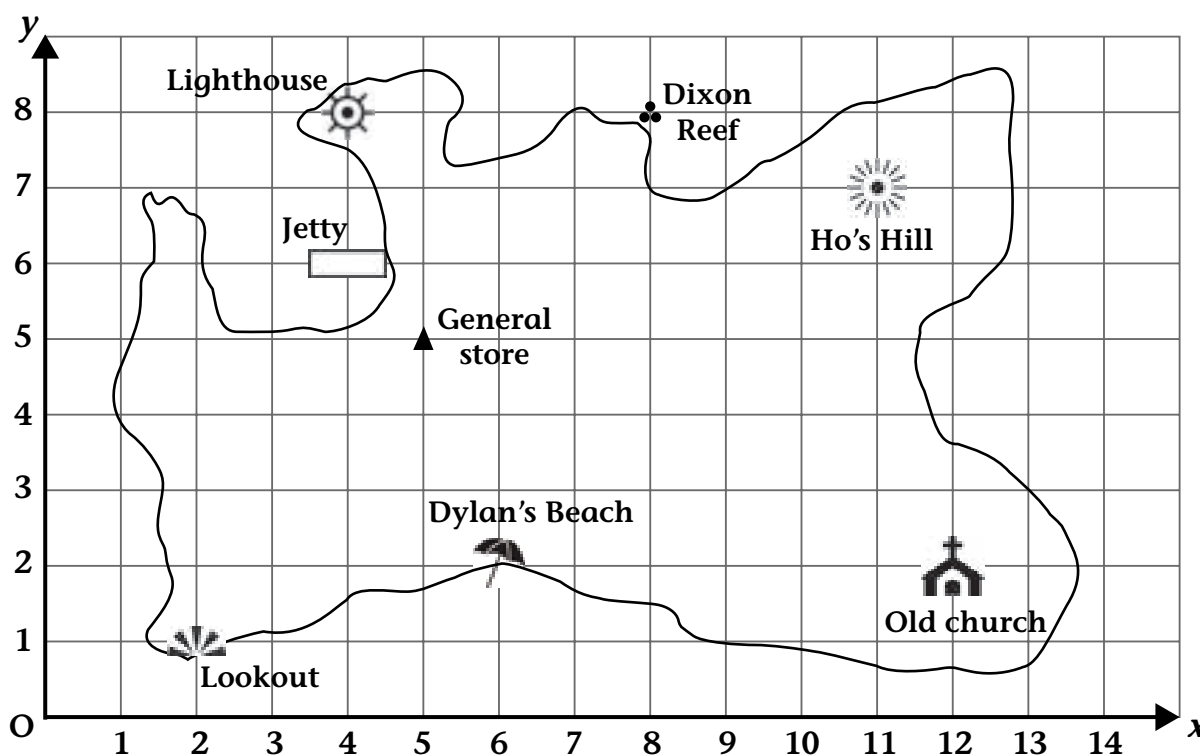


b Draw the two diagonals AC and BD on the diagram.

Where do the diagonals intersect? (Where do they cut?) _____

4 Melinda and Chris spent their holidays at Peaceful Island.

This is a map of the island.



What are the co-ordinates for:

a the general store? _____

b the lookout? _____

c the lighthouse? _____

d Dylan's Beach? _____

e Dixon Reef? _____

f Ho's Hill? _____

g the jetty? _____

h the old church? _____

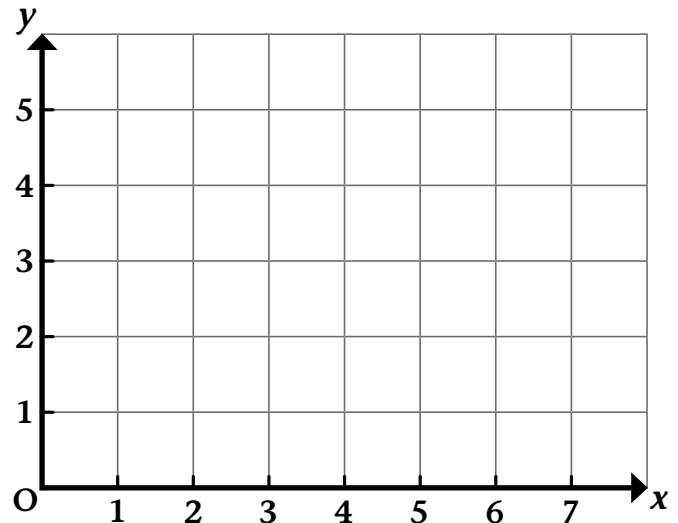
4.3 Locating points

Each ordered pair locates a point on a number plane.
We mark the point with a cross or a large dot.

- 1** Plot the following points on the number plane provided.

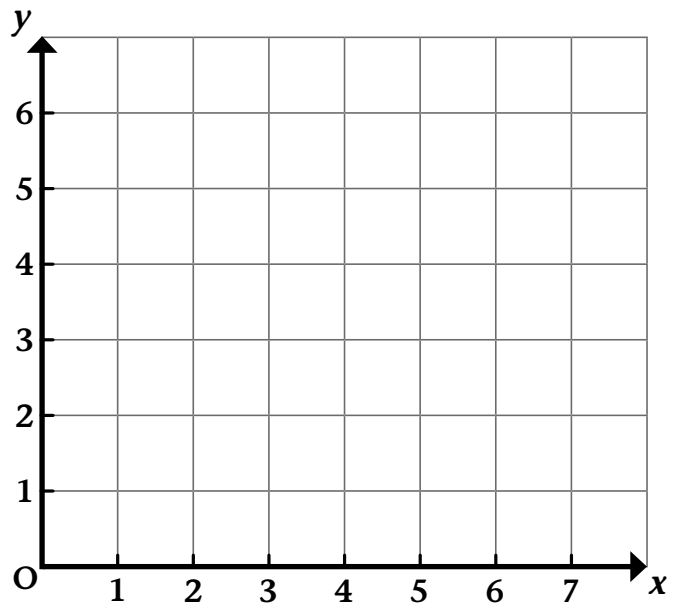
(2, 1), (3, 1), (4, 1), (5, 1),
(6, 1), (6, 2), (6, 3), (6, 4),
(5, 4), (4, 4), (3, 4), (2, 4),
(2, 3), (2, 2)

All these points lie on
a plane shape.
Name the plane shape.



- 2** Plot the following points
on the number plane provided.
Label each point with
the given letter.

A (2, 0)
B (4, 0)
C (6, 2)
D (6, 4)
E (4, 6)
F (2, 6)
G (0, 4)
H (0, 2)



Join the points in order, A to B
to C and so on. Then join H to A.

Name this plane shape.



4.3 Locating points

CONTINUED

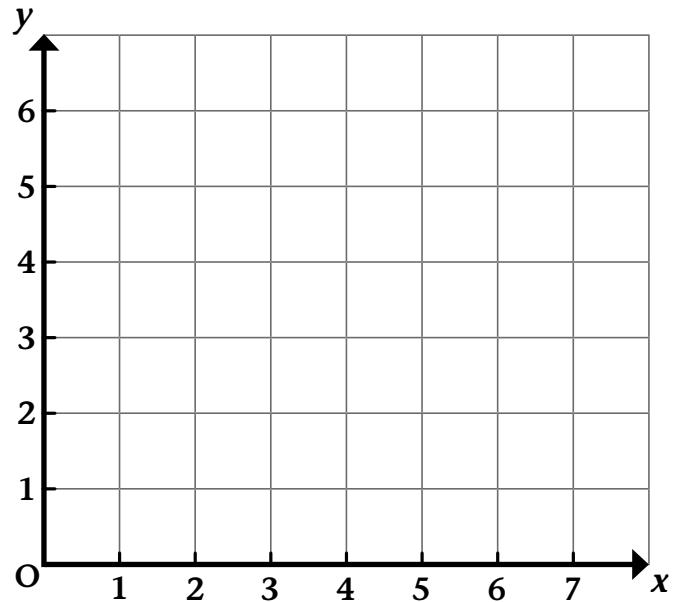
- 3 Plot the following points on the number plane provided. Join them in order with straight lines.

$(1, 2) \rightarrow (2, 2) \rightarrow (3, 2) \rightarrow (4, 2) \rightarrow$
 $(5, 2) \rightarrow (6, 2) \rightarrow (5, 1) \rightarrow (4, 1) \rightarrow$
 $(3, 1) \rightarrow (2, 1) \rightarrow (1, 2)$

Draw a line joining
 $(3, 2)$ and $(3, 5)$.

Join $(3, 5)$ and $(1, 2)$.

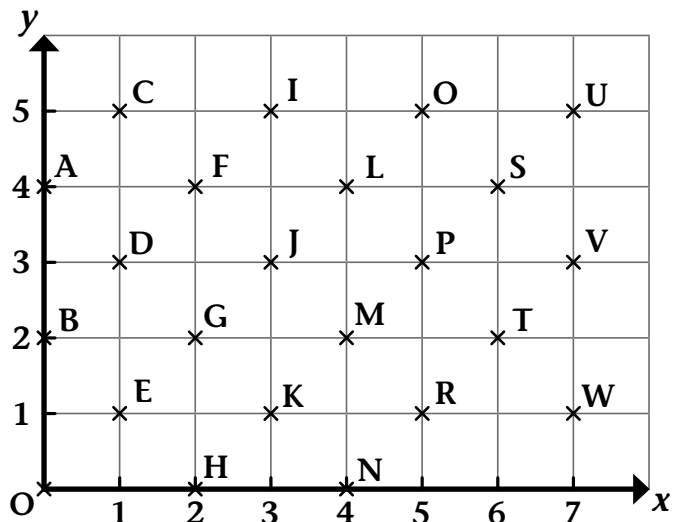
Join $(3, 5)$ and $(6, 2)$.



- 4 What did the tiger call the two tourists on bicycles?

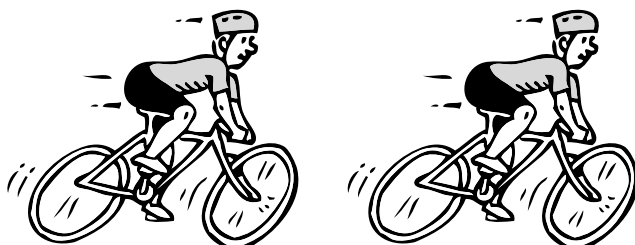
To answer this question, first find the location of each ordered pair listed below the number plane.

Write the letter labelling the point in the box below the ordered pair.



$(4, 2)$ $(1, 1)$ $(0, 4)$ $(4, 4)$ $(6, 4)$ $(5, 5)$ $(4, 0)$ $(7, 1)$ $(2, 0)$ $(1, 1)$ $(1, 1)$ $(4, 4)$ $(6, 4)$

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Teacher's notes

OBJECTIVES

- Interpret the number plane formed from the intersection of a horizontal x -axis and a vertical y -axis.
- Identify the origin and the four quadrants of the number plane.
- Read, plot and name ordered pairs on a number plane, using two or four quadrants.

PRIOR KNOWLEDGE

Students should be able to locate points on a grid using an ordered pair of co-ordinates.

They should be able to place positive and negative numbers in order along a number line.

BACKGROUND

In this unit, the notion of locating position established in Units 3 and 4 is further developed to include both positive and negative numbers as co-ordinates. The use of the four quadrant number plane is introduced.

STARTER ACTIVITY

Students could be given a sheet of grid paper with the axes drawn for the four quadrants, but not labelled. The teacher reads the instructions; for example: *Start where the axes meet. Go 3 units to the left, then down 4 units. Mark this point.*

The answer can be displayed on an interactive whiteboard or overhead projector. The teacher may wish to discuss movement to the left as the negative direction.

ACTIVITY SHEETS

5.1 Using two quadrants

This activity sheet introduces the second quadrant of the number plane. Students are asked to match points and their co-ordinates.

Answers

- 1 A (5, 2) B (4, 6) C (2, 0) D (0, 3)
E (-2, 4) F (-4, 2) G (-5, 5) H (-7, 3)
- 2 a D b G c I d F e A f H
g M h E i B j C k L l K
- 3 When it honks

5.2 Plotting points in two quadrants

In this activity sheet, students receive practice with plotting points in the first and second quadrants of a number plane.

Answers

- 2 b a hexagon

5.3 Using four quadrants

This activity sheet introduces the third and fourth quadrants of the number plane. Teachers should point out that the four quadrants are numbered in an anti-clockwise direction.

Students are asked to match points and their co-ordinates. They are also asked to identify the quadrants in which the x and y co-ordinates are positive or negative.

Answers

- 1 A (5, 2) B (3, 1) C (2, 4) D (0, 3)
E (-3, 2) F (-6, 4) G (-5, 0) H (-6, -2)
I (-5, -4) K (0, -3) L (2, -1) M (3, -4)
N (6, -2) P (-3, -1)
- 2 a A (2, 3) D (5, 2)
b B (-4, 2) F (-2, 1) H (-3, 4)
c C (-4, -1) G (-2, -3)
d E (1, -2) I (5, -1) K (3, -3)

3

Quadrant	x co-ordinate	y co-ordinate
First	positive	positive
Second	negative	positive
Third	negative	negative
Fourth	positive	negative

5.4 Plotting points in four quadrants

In this activity sheet, students receive practice plotting points in all four quadrants of a number plane.

Answers

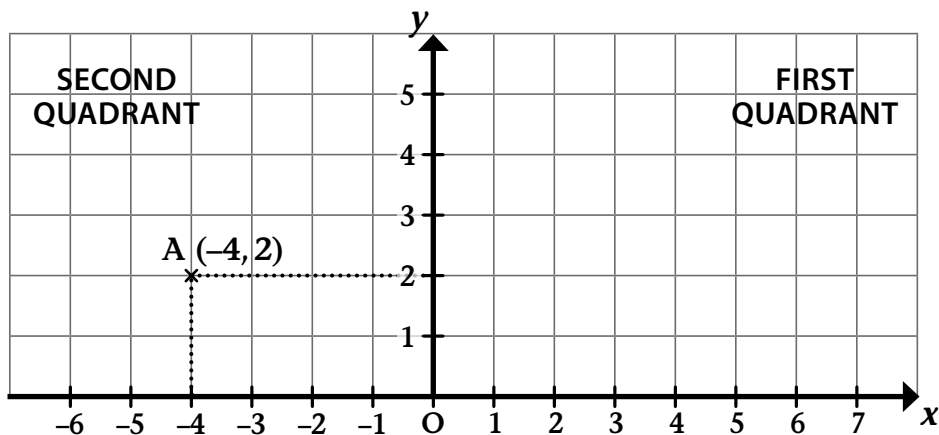
- 2 b an octagon

EXTENSION IDEAS

The class could play a game of *Battleships*. Each student could be given a copy of one of the number planes provided on page 63. Each player has one aircraft carrier (covering 6 points in a straight line), one battleship (4 points), two frigates (3 points) and four destroyers (2 points). The teacher could call out the coordinates where the bombs or shells land and students cross these points off on their number planes. Alternatively, students could take turns to call out the coordinates of the points. A hit will sink any ship. The last student with a ship intact is the winner.

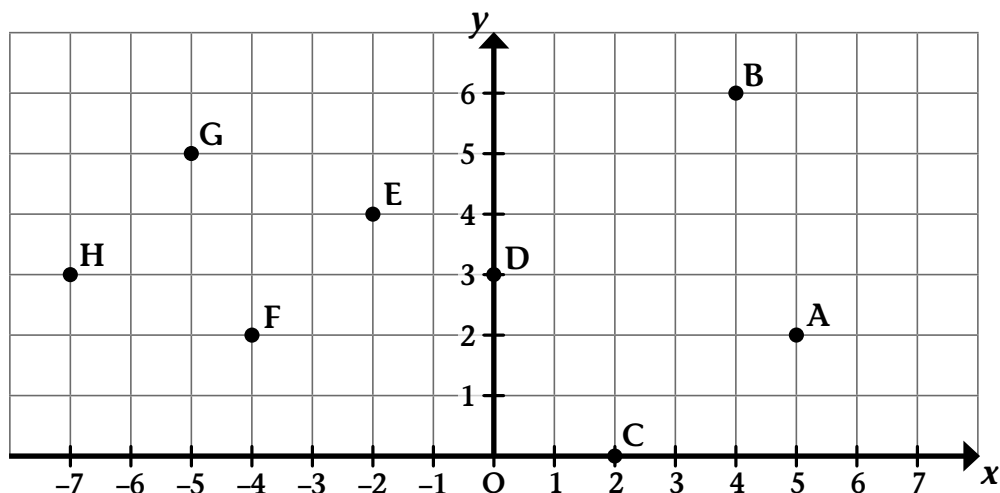
5.1 Using two quadrants

Casey wanted to extend a number plane to the left.
She placed the vertical axis in the middle of the number plane.
She extended the horizontal axis (x -axis) to the left using negative numbers.



The region to the left of the y -axis is called the second quadrant.
Points in the second quadrant have a negative x co-ordinate.
Their y co-ordinate is positive.
The point A has co-ordinates $(-4, 2)$.

- 1 Points A to H have been labelled on this number plane.
Write the co-ordinates for each point.



A _____ B _____
C _____ D _____
E _____ F _____
G _____ H _____

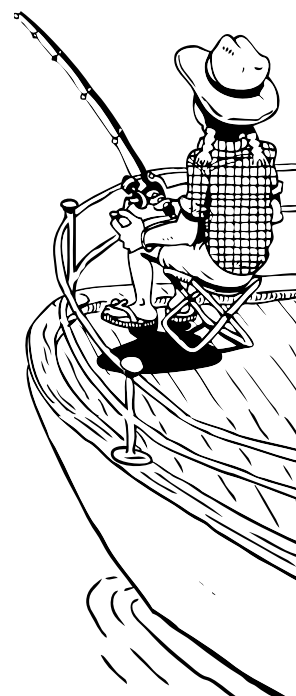
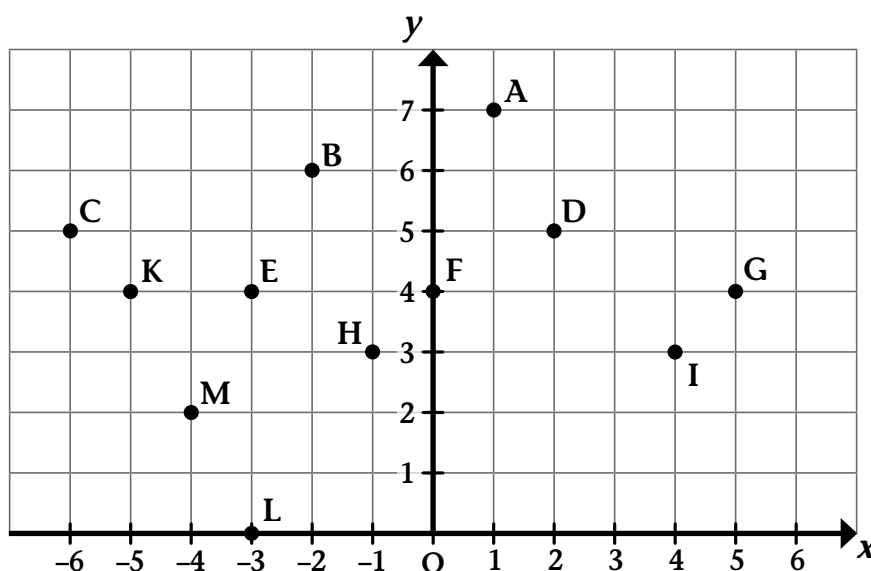


5.1 Using two quadrants

CONTINUED

2 Name the points with these co-ordinates.

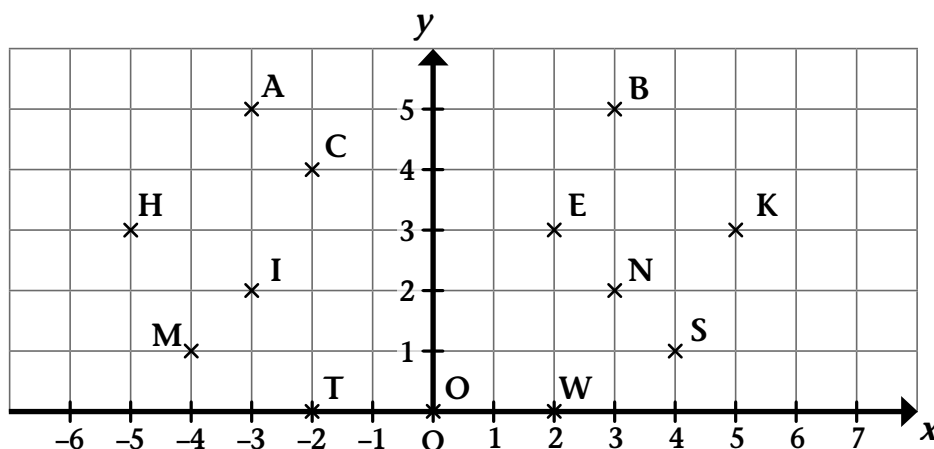
- | | | |
|-----------------|-----------------|-----------------|
| a (2, 5) _____ | b (5, 4) _____ | c (4, 3) _____ |
| d (0, 4) _____ | e (1, 7) _____ | f (-1, 3) _____ |
| g (-4, 2) _____ | h (-3, 4) _____ | i (-2, 6) _____ |
| j (-6, 5) _____ | k (-3, 0) _____ | l (-5, 4) _____ |



3 When is a goose like a truck?

To answer this question, first find the location of each ordered pair listed below the number plane.

Write the letter labelling the point in the box below the ordered pair.



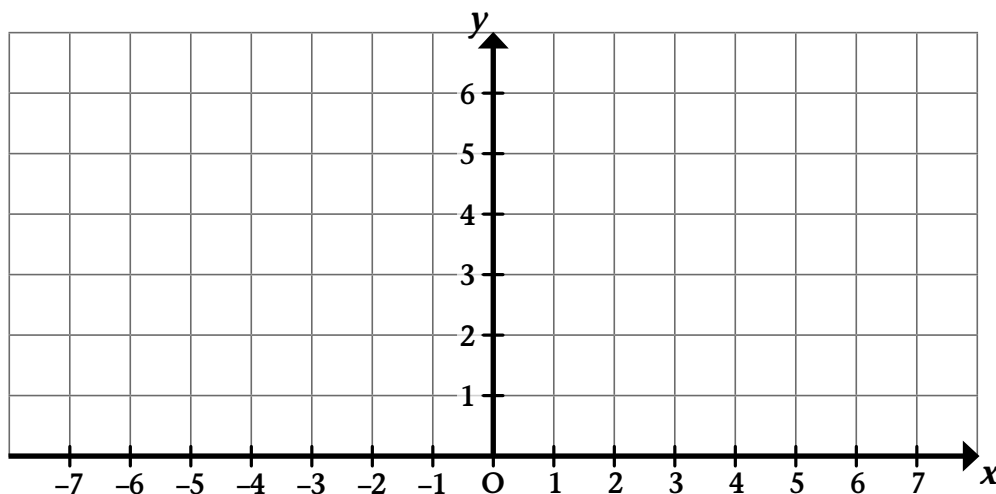
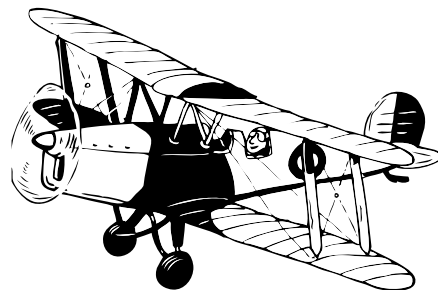
(2, 0) (-5, 3) (2, 3) (3, 2) (-3, 2) (-2, 0) (-5, 3) (0, 0) (3, 2) (5, 3) (4, 1)

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5.2 Plotting points in two quadrants

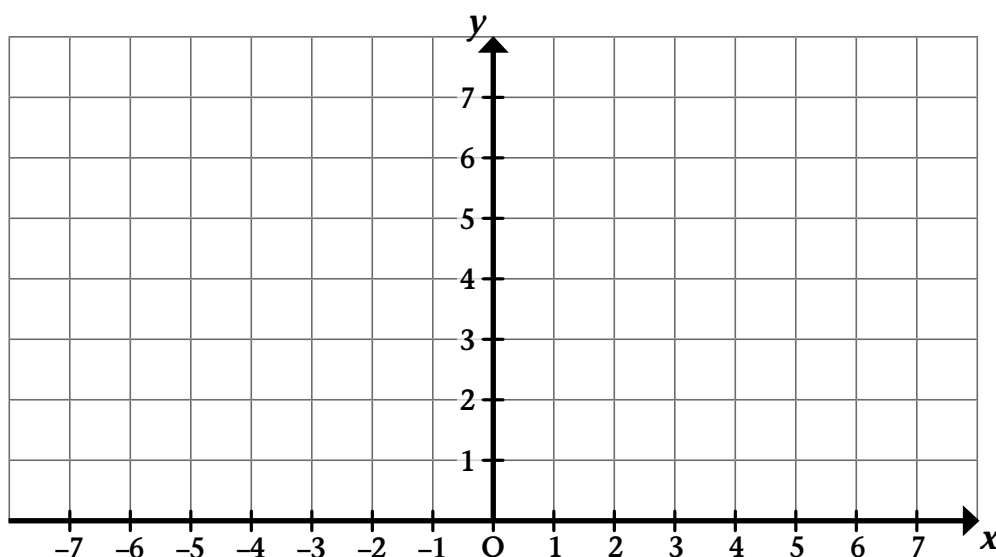
- 1** Plot these points on the number plane provided.
Label each point with its co-ordinates.

A (2, 3) B (-2, 3) C (-3, 2) D (5, 2)
E (-5, 2) F (-2, 5) G (3, 6) H (-3, 6)
I (0, 5) J (6, 5) K (-6, 5) L (-7, 0)



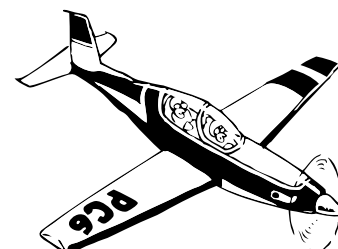
- 2** Plot these points on the number plane provided.

A (2, 2) B (2, 5) C (-1, 7) D (-4, 5) E (-4, 2) F (-1, 0)



Use straight lines to join the points in order.
(Join A to B to C and so on.) Then join F to A.

Name the plane shape that is formed.



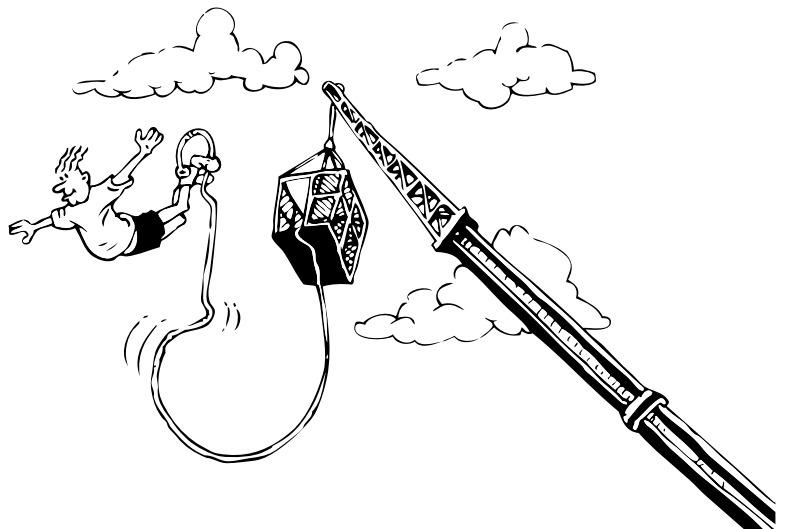
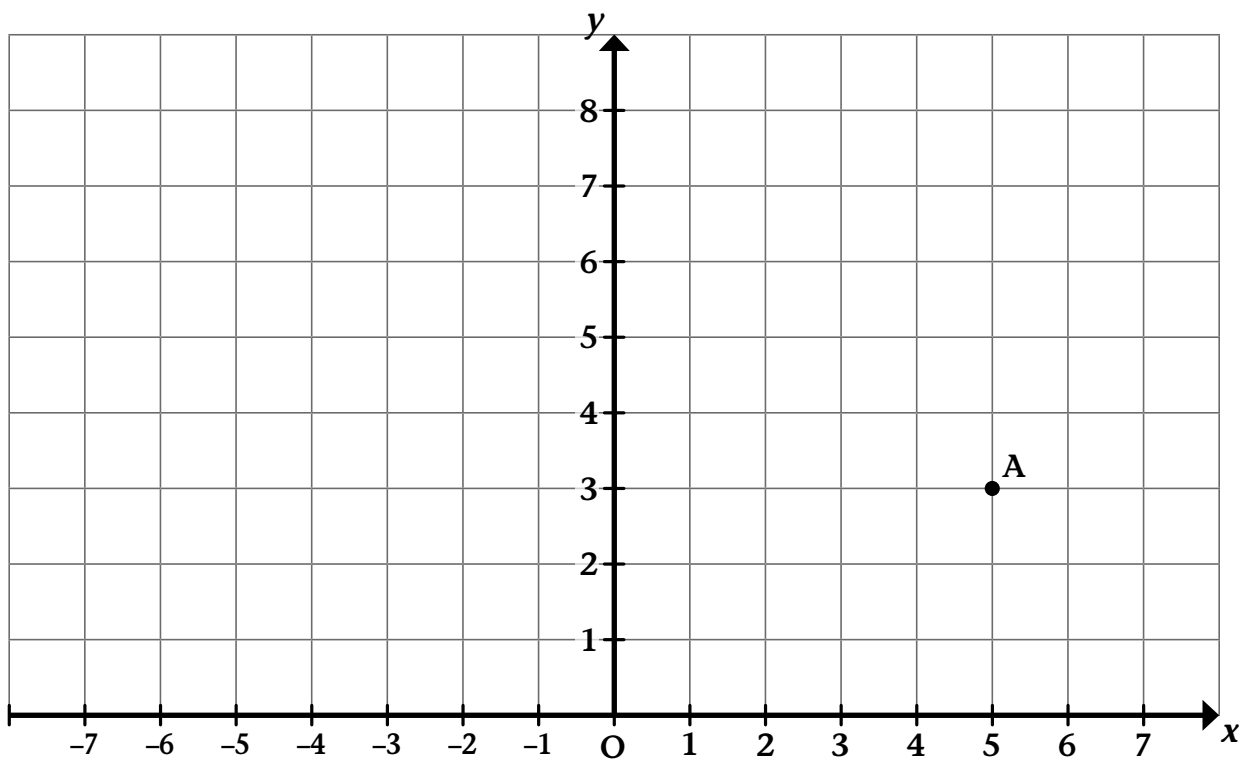
5.2 Plotting points in two quadrants

CONTINUED

- 3** Plot the following points on the number plane provided.
Join the points in order with straight lines.

A (5, 3) → B (6, 3) → C (5, 4) → D (1, 6) → E (2, 5) →
F (0, 6) → G (1, 5) → H (-2, 4) → I (-4, 3) → J (-6, 1) →
K (-5, 3) → L (-6, 5) → M (-4, 3) → N (1, 1) → O (0, 0) →
P (2, 1) → Q (1, 0) → R (5, 2) → S (6, 3)

The starting point A is plotted for you.



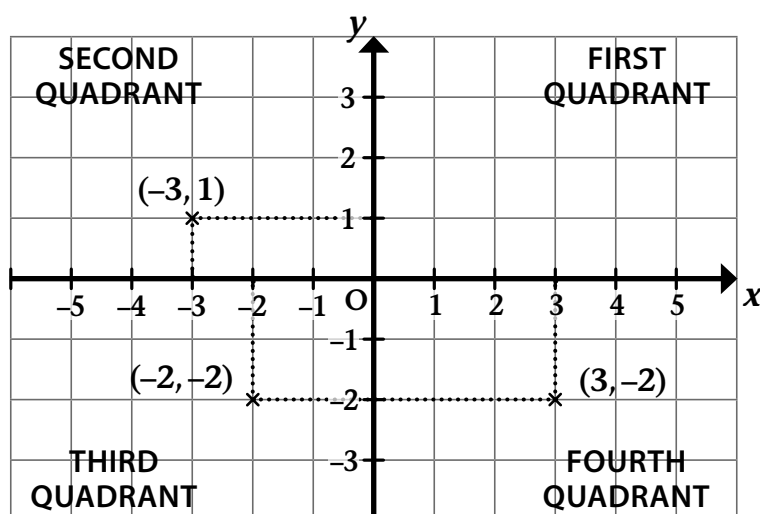
5.3 Using four quadrants

Shaun wanted a number plane that extended up and down, as well as to the left and right.

He used two axes that crossed in the middle of the grid.

The axes divide this number plane into **four quadrants**.

The two axes meet at the **origin** (0, 0).



Points in the second or third quadrant have a **negative** x co-ordinate.

Points in the third or fourth quadrant have a **negative** y co-ordinate.

1 Write the co-ordinates of the points shown in this number plane.

A _____

B _____

C _____

D _____

E _____

F _____

G _____

H _____

I _____

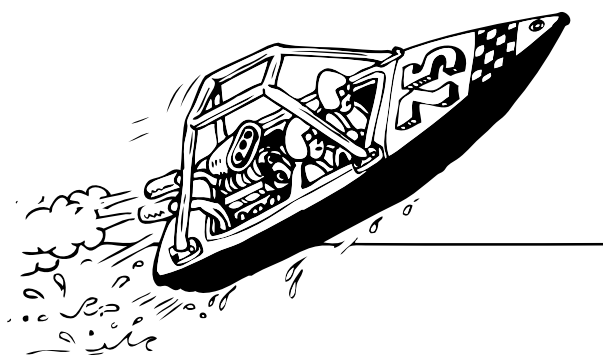
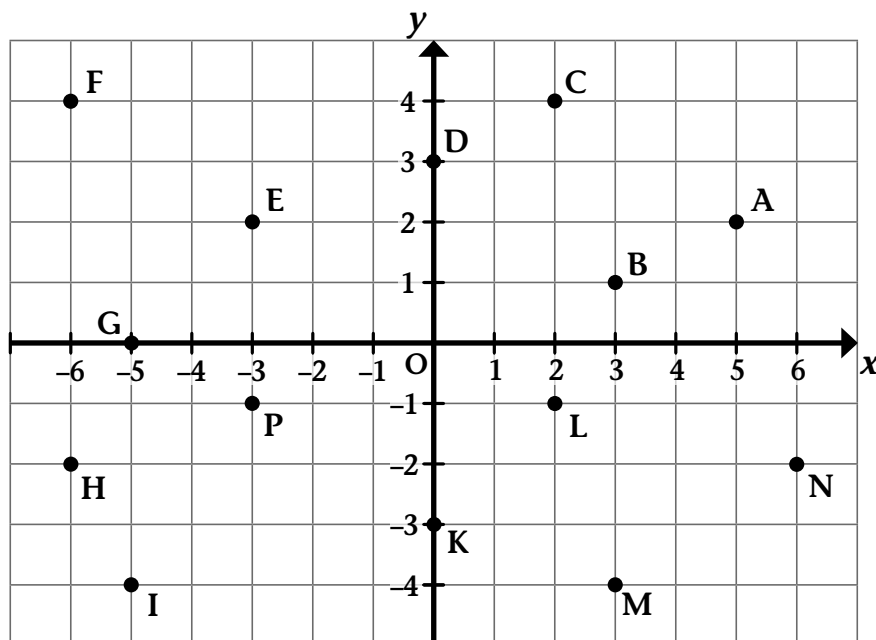
K _____

L _____

M _____

N _____

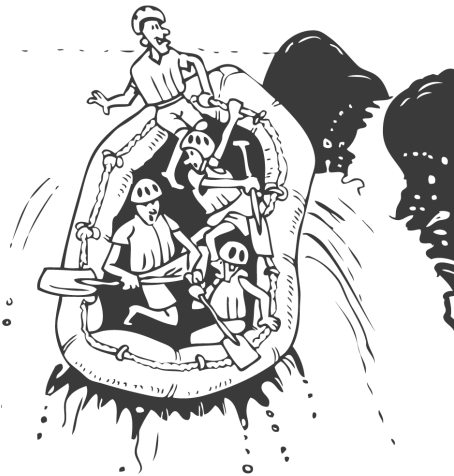
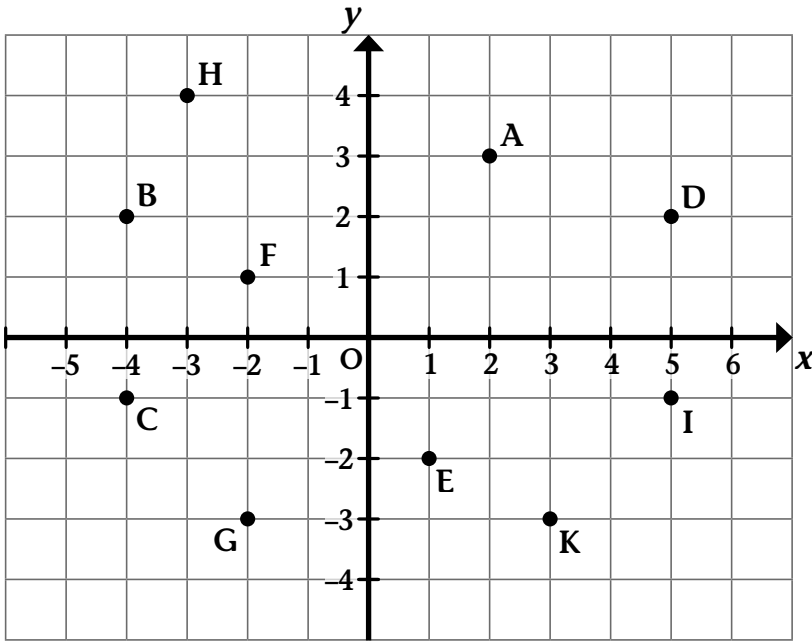
P _____



5.3 Using four quadrants

CONTINUED

2 Ten points have been labelled on this number plane.



List the points and their co-ordinates in:

- a the first quadrant. _____
- b the second quadrant. _____
- c the third quadrant. _____
- d the fourth quadrant. _____

3 Complete this table.

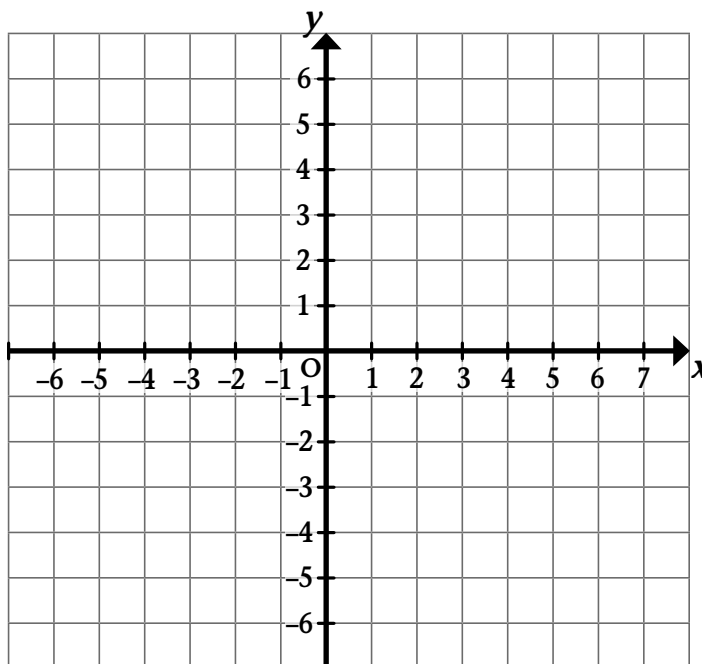
Quadrant	x co-ordinate	y co-ordinate
First	positive	positive
Second	negative	
Third		negative
Fourth		



5.4 Plotting points in four quadrants

- 1** Plot these points on the number plane provided.
Label each point with its co-ordinates.

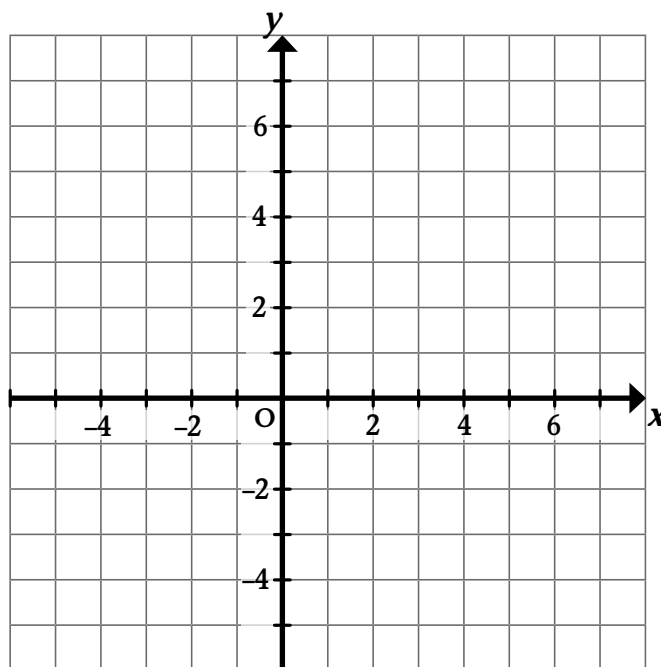
- A (4, 1)
- B (6, 5)
- C (-2, 2)
- D (-4, 6)
- E (-5, 0)
- F (-3, -4)
- G (-4, -2)
- H (0, -1)
- I (2, -3)
- J (6, -5)



- 2** Plot the following points on the number plane provided.
Use straight lines to join the points in order (A to B to C and so on).

A (5, 3) → B (3, 5) → C (-1, 5) → D (-3, 3) → E (-3, -1) →
F (-1, -3) → G (3, -3) → H (5, -1) → A (5, 3)

What plane shape
have you drawn?

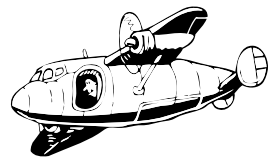
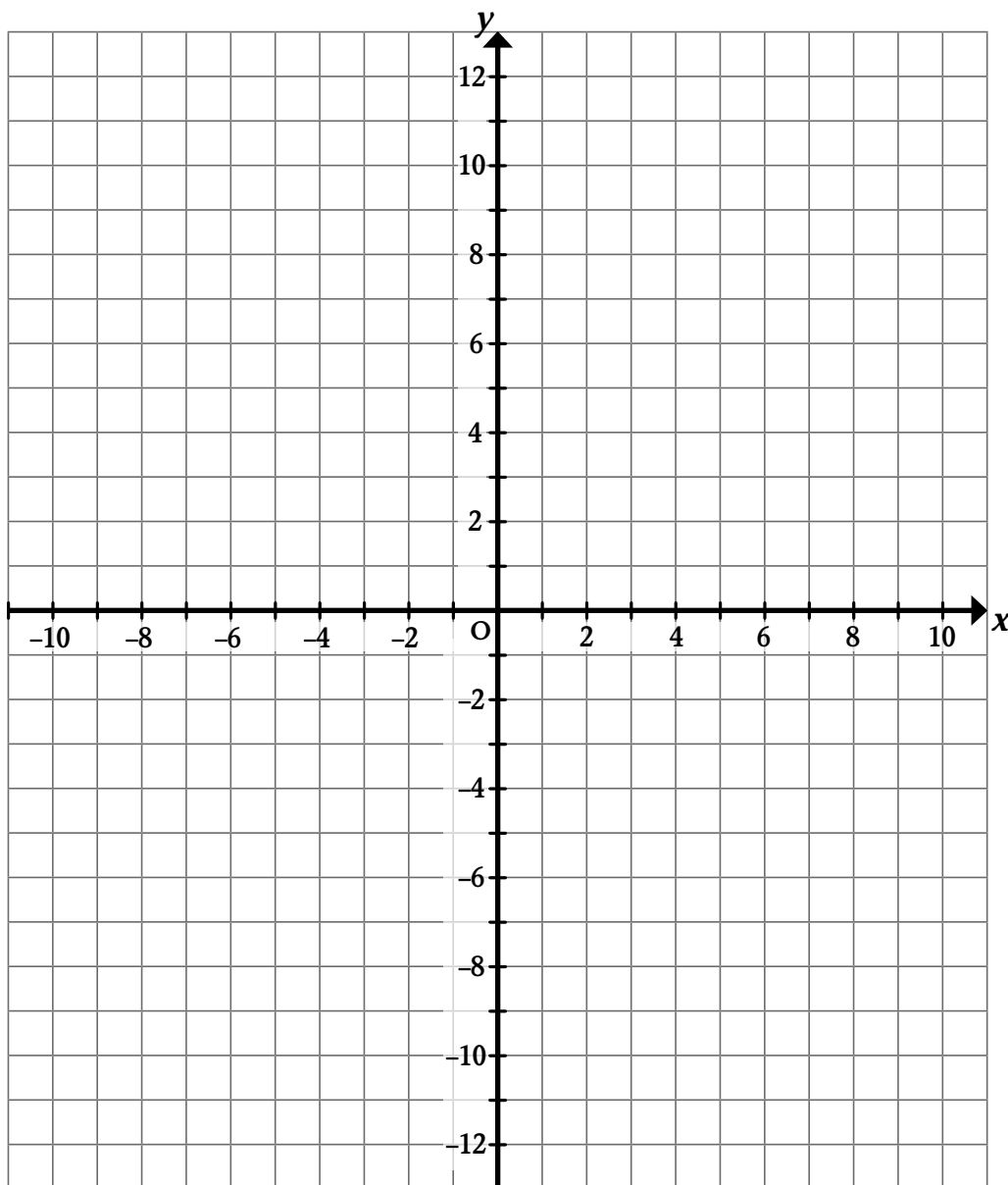


5.4 Plotting points in four quadrants

CONTINUED

- 3** Plot the following points on the number plane provided.
Join the points in order with straight lines.

A (8, 6) → B (-4, 6) → C (-10, 3) → D (-6, 6) → E (-6, 8) →
 F (-8, 10) → G (10, 10) → H (8, 8) → I (8, -4) → J (6, -6) →
 K (6, -10) → L (0, -10) → M (0, -8) → N (-6, -8) → O (-6, -6) →
 P (-4, -6) → Q (-4, -5) → R (-6, -5) → S (-6, -4) → T (-9, -4) →
 U (-8, -2) → V (-6, 0) → W (-4, 0) → X (-6, 1) → Y (-6, 5)



Teacher's notes

OBJECTIVES

- Graph points on a number plane from a table of values.
- Complete a table of values for a linear function then graph the ordered pairs on a number plane.
- Graph two intersecting lines on the same set of axes and read the co-ordinates of their point of intersection.
- Recognise positive and negative gradients from a graph.
- Use a right-angled triangle drawn between two points on a number plane to calculate the gradient of the interval joining the two points.

PRIOR KNOWLEDGE

Students should be able to perform the basic operations with whole numbers using mental arithmetic or a calculator. They should be able to complete a table of values for a linear relationship by substituting values for one of the pronumerals.

BACKGROUND

The process of drawing horizontal and vertical lines to find the gradient is applied in a variety of other situations; for example, finding angles of elevation or depression and plotting courses using compass bearings.

STARTER ACTIVITY

Revise tables of values. Give students a pattern built using toothpicks. Ask them to count the number of toothpicks used in each step and write their answers in a table.

Alternatively, give students the algebraic rule and ask them to complete a table of values.

ACTIVITY SHEETS

6.1 Using a table of values

In this activity sheet, students graph linear relationships created from number patterns and tables of ordered pairs.

Students will need a copy of page 63 (blank number planes) to complete questions 3, 4 and 5. Alternatively, teachers may prefer to have students use a sheet of blank grid paper (page 64) and draw their own axes.

Answers

- yes
- a (1, 1) (2, 4) (3, 7) (4, 10)
- (-1, 0) (0, 1) (1, 2) (2, 3)
- (-2, 5) (0, 3) (2, 1) (4, -1)
- (-2, 0) (0, 2) (1, 3) (3, 5)

6.2 Graphs from equations A

In this activity sheet, students use a table of values to graph a simple linear equation. In question 1, the equation is based on a pattern of toothpicks. In questions 2 and 3 students complete a table of values and use these ordered pairs to graph simple linear equations.

Answers

2 a

x	-3	0	2
y	-1	2	4

3 a

x	-1	0	2
y	-2	0	4

6.3 Graphs from equations B

In this activity sheet, students complete a table of values and use these ordered pairs to graph simple linear equations. Students will need two copies of page 63 (blank number planes) to complete question 3.

Answers

1 a

x	-3	0	2
y	4	1	-1

b

x	-2	0	1
y	-3	1	3

Teacher's notes

2 a

x	0	2	4
y	-3	1	5

c

x	-1	0	2
y	-4	-1	5

3 a

x	-3	0	2
y	1	4	6

c

x	-1	0	2
y	3	0	-6

e

x	-2	0	1
y	-4	2	5

b

x	-2	0	1
y	4	0	-2

d

x	-2	0	2
y	5	1	-3

b

x	-2	0	3
y	-5	-3	0

d

x	-4	-2	1
y	-5	-1	5

f

x	-3	0	2
y	5	-1	-5

6.4 Intersecting lines

In this activity sheet, students use tables of values to draw the graphs of two linear equations on the same number plane. They then read the co-ordinates of the point of intersection.

Students will need a copy of page 63 (blank number planes) to complete question 3.

Alternatively, teachers may prefer to have students use a sheet of blank grid paper (page 64) and draw their own axes.

Answers

1 $y = x + 2$

x	-2	0	3
y	0	2	5

$y = 2x$

x	-2	0	3
y	-4	0	6

The point of intersection is (2, 4)

2 $y = x - 2$

x	-2	0	3
y	-4	-2	1

$y = 2 - x$

x	-2	0	3
y	4	2	-1

The point of intersection is (2, 0)

3 a (2, 2) b (1, 2) c (1, 3) d (-2, -3)

6.5 The gradient of a line A

In this activity sheet, students determine whether a line has positive or negative gradient by following the line left to right – if the line slopes up, the gradient is positive; if the line slopes down, the gradient is negative.

Answers

- 1 a positive b negative c positive
d positive e negative f negative

6.6 The gradient of a line B

In this activity sheet, students form a right-angled triangle by drawing a vertical line from the higher point and a horizontal line from the lower point. They then use this triangle and the rule

$$\text{GRADIENT} = \frac{\text{VERTICAL RISE}}{\text{HORIZONTAL RUN}}$$

to calculate the gradient of the line.

1 a 3 b 2 c -1 d $\frac{1}{2}$ e -1 f $-\frac{1}{2}$
g $\frac{2}{3}$ h $-\frac{5}{4} = -1\frac{1}{4}$

6.7 Calculating the gradient

In this activity sheet, students use a graph to calculate the gradient of the line joining two points, using a right-angled triangle and the rule:

$$\text{GRADIENT} = \frac{\text{VERTICAL RISE}}{\text{HORIZONTAL RUN}}$$

In question 2, they are encouraged to calculate the gradient without drawing a graph.

Answers

1 a 1 b $\frac{2}{3}$ c -1 d $-\frac{1}{2}$
2 a $\frac{3}{4}$ b -1 c $\frac{1}{3}$ d $-\frac{5}{4} = -1\frac{1}{4}$
e $\frac{1}{2}$ f $-\frac{1}{2}$

EXTENSION IDEAS

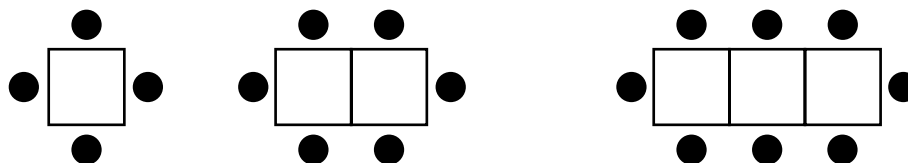
Ask students to check if a given point lies on a line by substituting the co-ordinates into the equation of the line.

Students could use a graphics calculator and spreadsheet software to graph and compare a range of linear relationships, including intersecting lines and parallel lines.

They could investigate which term in a linear equation (of the form $y = mx + b$) affects the slope of a graph, making the gradient positive or negative. The teacher may wish to introduce the terms *increasing* and *decreasing* to describe the graph.

6.1 Using a table of values

A restaurant has square tables. Each table seats one person on each side. Larger groups are seated by moving tables together.



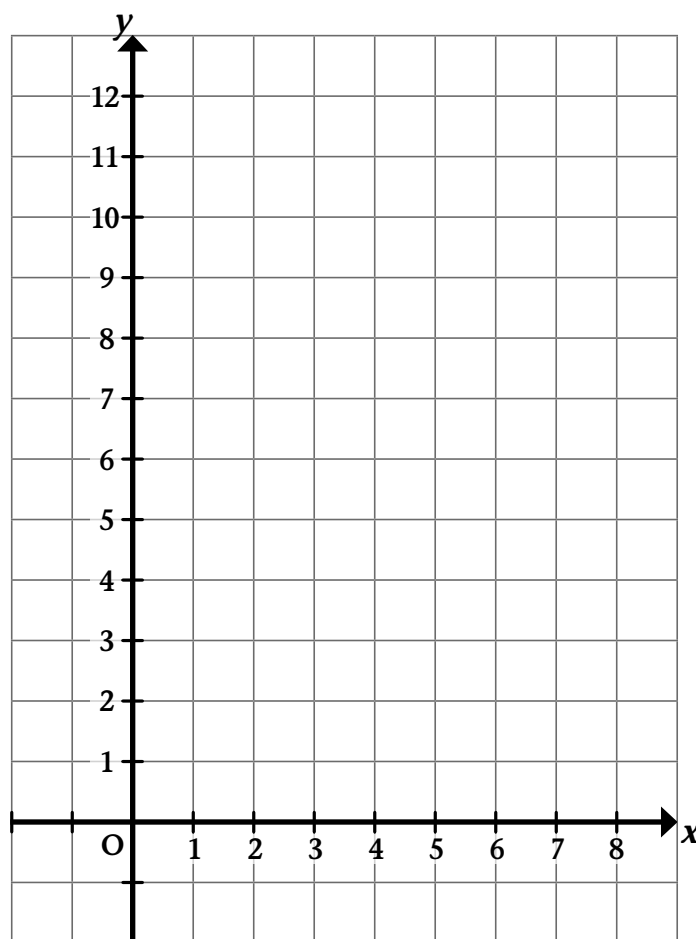
The different arrangements are shown in this table.

Number of tables (x)	1	2	3	4	5
Number of people (y)	4	6	8	10	12

Thuan drew a graph to show the pattern.

Each vertical pair of numbers gives a point on the graph.

- The ordered pairs from the table are (1, 4), (2, 6), (3, 8), (4, 10) and (5, 12). Plot these points on the grid below.



Can you draw a straight line passing through these five points? _____

6.1 Using a table of values

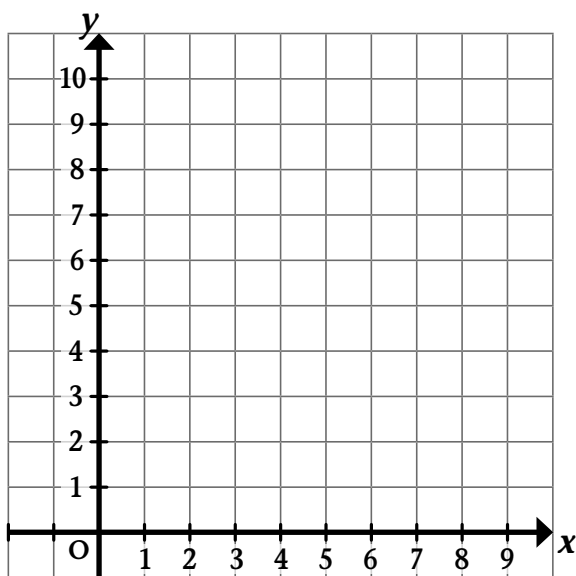
CONTINUED

- 2** Judy completed this table using the rule $y = 3x - 2$.

x	1	2	3	4
y	1	4	7	10

- a** List the ordered pairs shown in this table.

- b** Plot these points on the grid below.
(Check: They should lie on a straight line.)



For each table of values:

- a** list the ordered pairs shown in the table
b plot the points on a number plane. (They should lie on straight lines.)

3

x	-1	0	1	2
y	0	1	2	3

The ordered pairs are:

4

x	-2	0	2	4
y	5	3	1	-1

The ordered pairs are:

5

x	-2	0	1	3
y	0	2	3	5

The ordered pairs are:



6.2 Graphs from equations A

Marianne built this pattern using toothpicks.



The number of toothpicks she used in each step is given by the rule:

$y = 3x + 1$ y = number of toothpicks
 x = number of squares

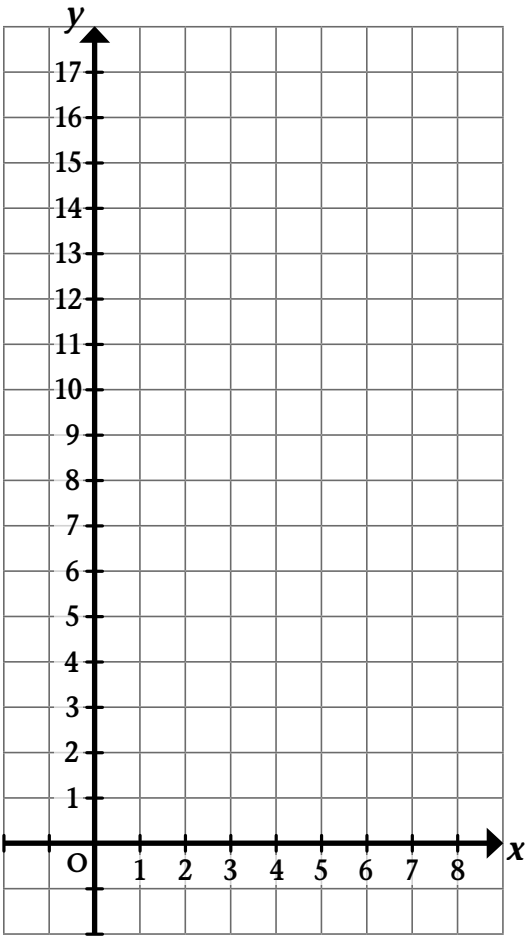
The pattern can also be shown in a table.

Number of squares (x)	1	2	3	4	5
Number of toothpicks (y)	4	7	10	13	16

This can be considered as the point (1, 4).

This is the point (3, 10).

- 1 The ordered pairs given in the table are: (1, 4), (2, 7), (3, 10), (4, 13) and (5, 16). Plot these points on the number plane. You can draw a straight line through these points. This line is the graph of $y = 3x + 1$.



6.2 Graphs from equations A

CONTINUED

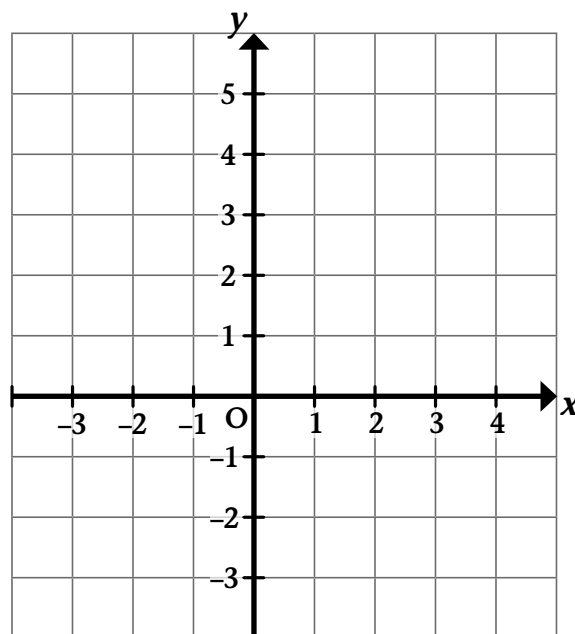
- 2** Matthew wanted to draw the graph of the equation $y = x + 2$. He chose three values to substitute for x : -3 , 0 and 2 .

- a** Complete the table of values for $y = x + 2$.

x	-3	0	2
y			

- b** Plot the ordered pairs from the table on this number plane.

Draw a straight line through these points to complete the graph of the line $y = x + 2$.

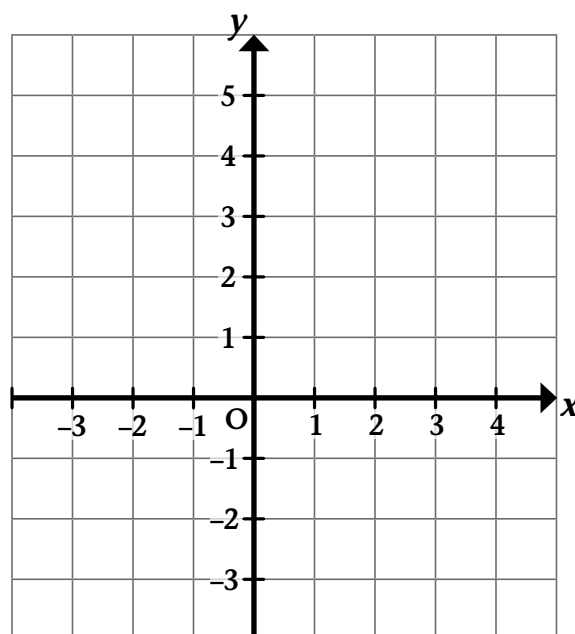


- 3 a** Complete this table of values for $y = 2x$

x	-1	0	2
y			

You can put any numbers you like in the top row.

- b** Draw the graph of $y = 2x$ on this number plane.



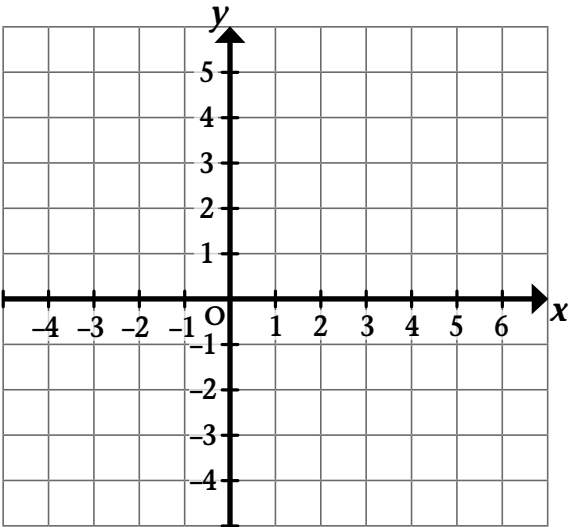
6.3 Graphs from equations B

1 Complete each table of values

a $y = 1 - x$

x	-3	0	2
y			

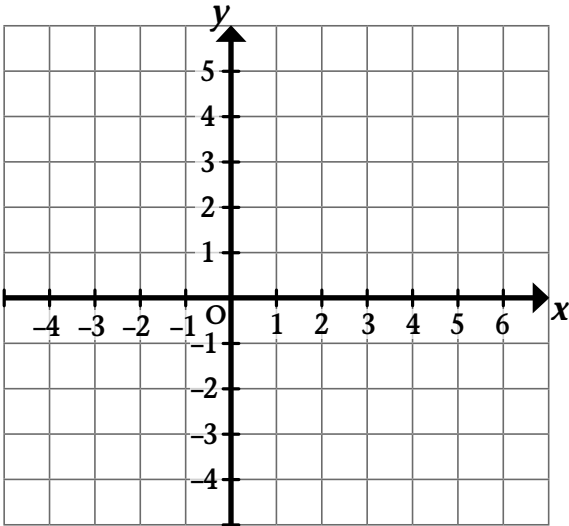
Draw the graph of each line on the number plane below.



b $y = 2x + 1$

x	-2	0	1
y			

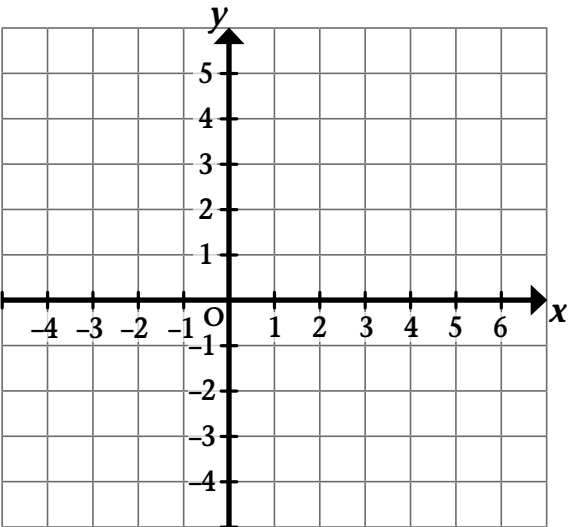
You can put any numbers you like in the top row.



2 Complete each table of values.
Then, draw the graph of each line on the number plane provided.

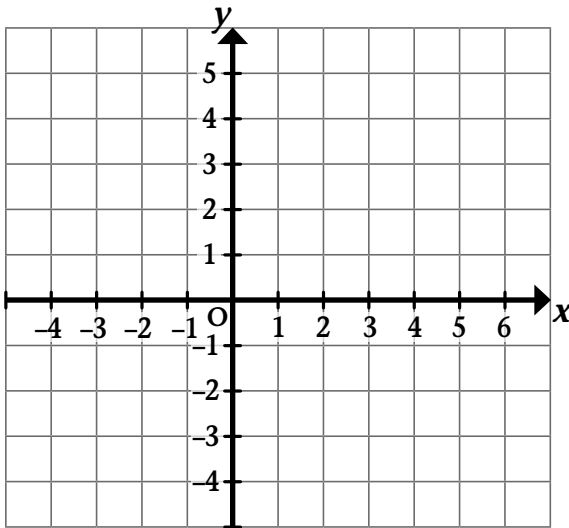
a $y = 2x - 3$

x	0	2	4
y			



b $y = -2x$

x	-2	0	1
y			

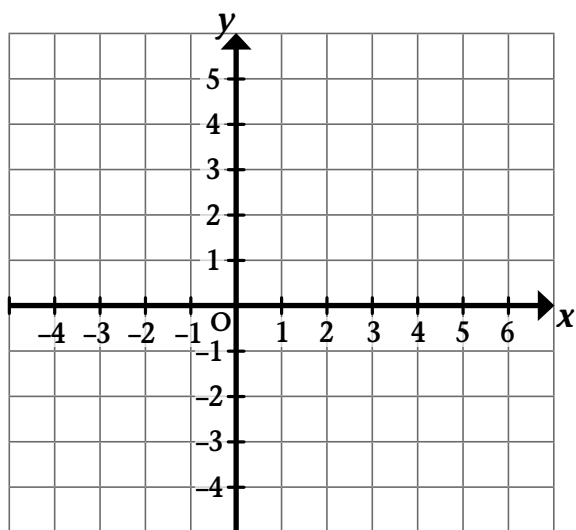


6.3 Graphs from equations B

CONTINUED

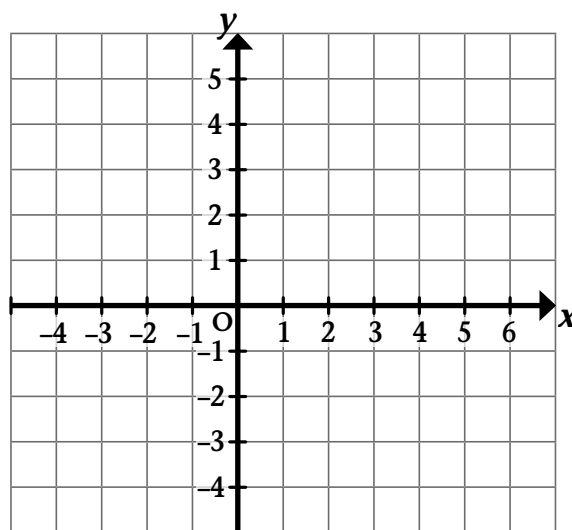
c $y = 3x - 1$

x	-1	0	2
y			



d $y = -2x + 1$

x	-2	0	2
y			



3 Complete each table of values.

a $y = x + 4$

x	-3	0	2
y			

b $y = x - 3$

x	-2	0	3
y			

c $y = -3x$

x	-1	0	2
y			

d $y = 2x + 3$

x	-4	-2	1
y			

e $y = 3x + 2$

x	-2	0	1
y			

f $y = -2x - 1$

x	-3	0	2
y			



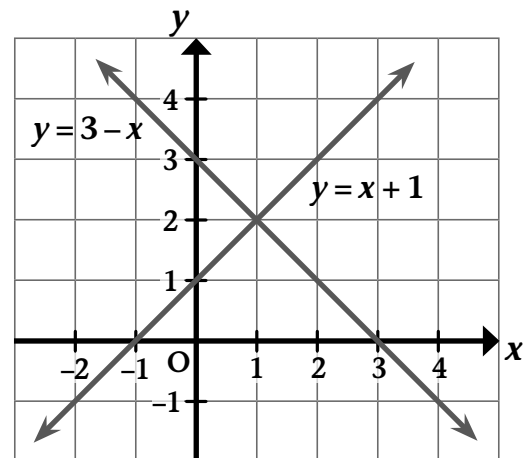
Draw the graph of each line on a number plane.

6.4 Intersecting lines

Danny graphed the lines $y = x + 1$ and $y = 3 - x$ on the same number plane.

The two lines cut at the point $(1, 2)$.

This is called the **point of intersection** of the two lines.



- 1 **a** Graph the lines $y = x + 2$ and $y = 2x$ on the same number plane.
- b** Write the co-ordinates of the point of intersection.

Step 1: Complete a table of values for each equation.

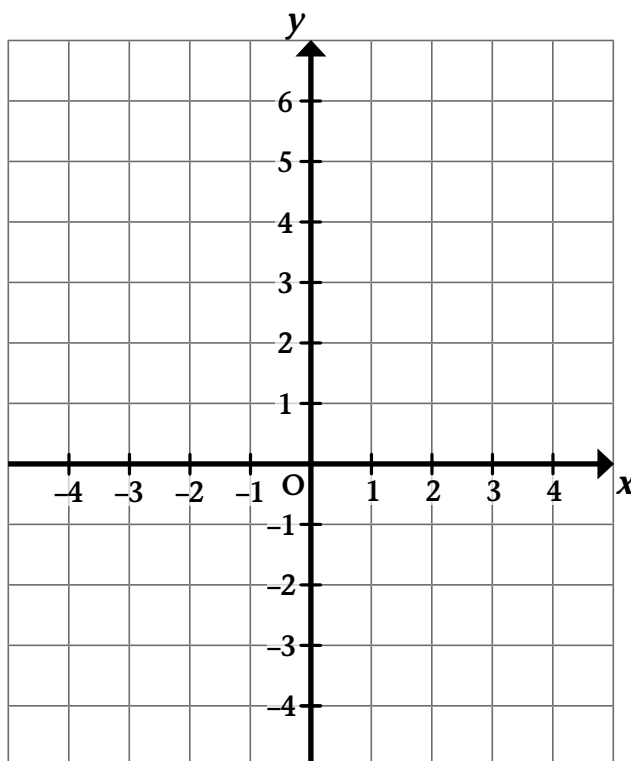
$$y = x + 2$$

x	-2	0	3
y			

$$y = 2x$$

x	-2	0	3
y			

Step 2: Graph the lines on the same number plane.



Step 3: Read the co-ordinates of the point of intersection (where they cut).

6.4 Intersecting lines

CONTINUED

- 2** Graph the lines $y = x - 2$ and $y = 2 - x$ on the same number plane. Write the co-ordinates of the point of intersection.

Step 1: Complete a table of values for each equation.

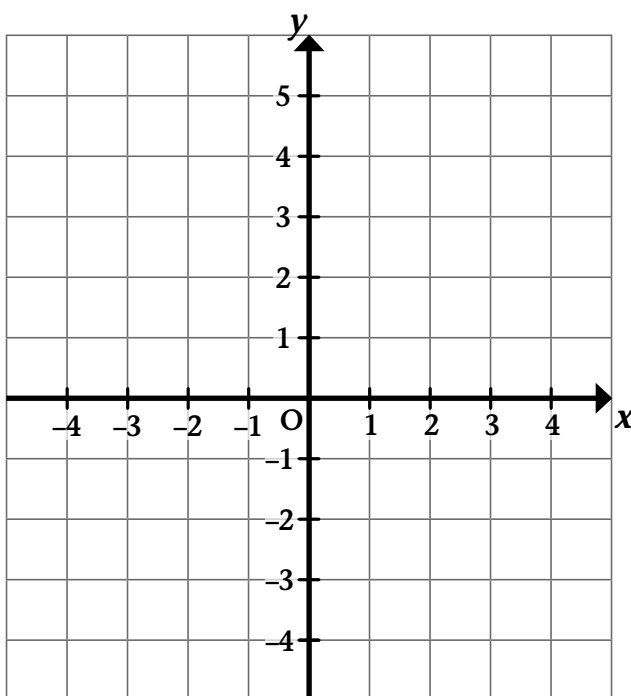
$$y = x - 2$$

x	-2	0	3
y			

$$y = 2 - x$$

x	-2	0	3
y			

Step 2: Graph the lines on the same number plane.



Step 3: The point of intersection is _____

- 3** Graph each pair of lines on the same number plane. Write the co-ordinates of their point of intersection. (You will need a separate number plane for each pair of lines.)

a $y = x$ and $y = 4 - x$ _____

b $y = x + 1$ and $y = 3 - x$ _____

c $y = 3x$ and $y = x + 2$ _____

d $y = 2x + 1$ and $y = x - 1$ _____

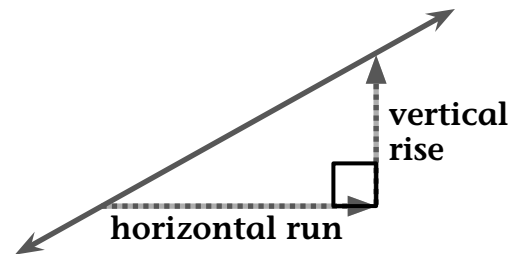


6.5 The gradient of a line A

Gradient is another word for the slope of a line.

The gradient is calculated using the rule:

$$\text{GRADIENT} = \frac{\text{VERTICAL RISE}}{\text{HORIZONTAL RUN}}$$



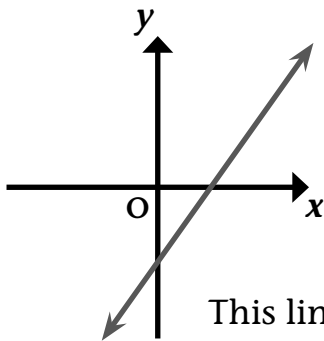
Follow the line from left to right.

Lines which **slope up** to the right have a **positive gradient**.

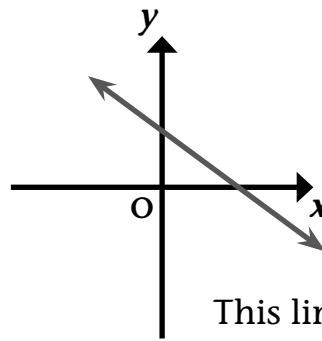
Lines which **slope down** to the right have a **negative gradient**.

The gradient of a **horizontal** line is **zero**.

EXAMPLES

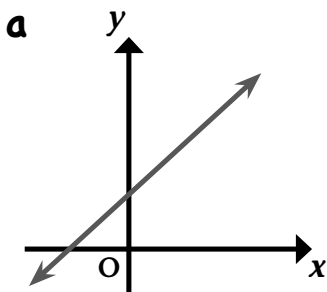


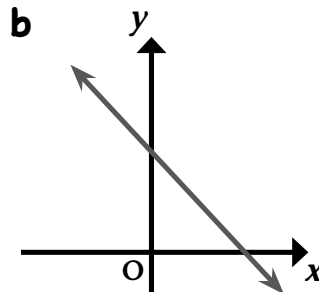
This line slopes up.
The gradient is **positive**.

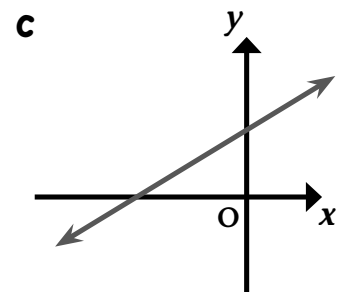


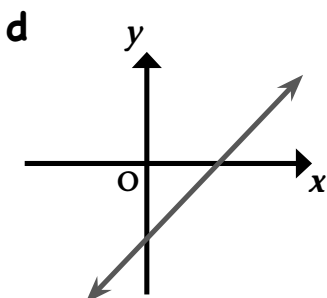
This line slopes down.
The gradient is **negative**.

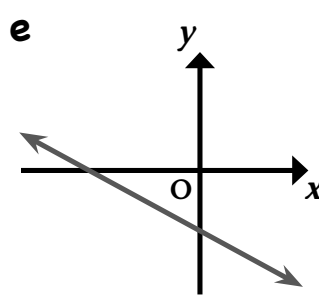
1 Are the gradients of these lines positive or negative?

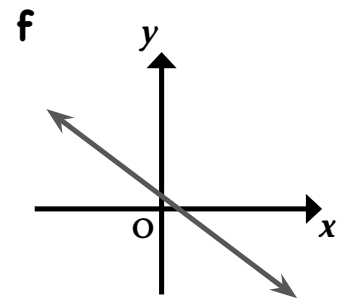












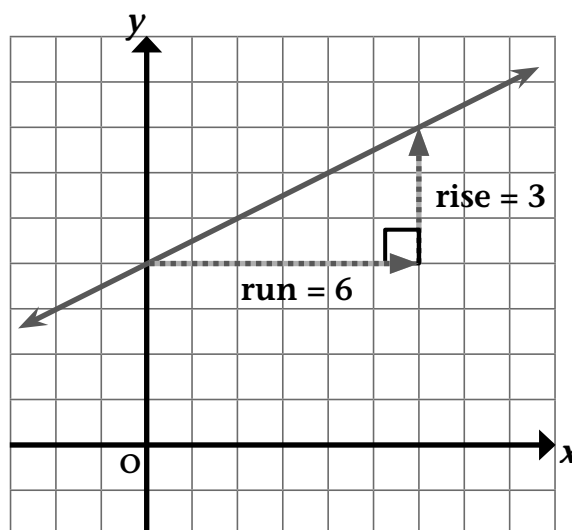
6.6 The gradient of a line B

The gradient is calculated using the rule:

$$\text{GRADIENT} = \frac{\text{VERTICAL RISE}}{\text{HORIZONTAL RUN}}$$

In this diagram:

$$\begin{aligned} \text{Gradient} &= \frac{\text{Vertical rise}}{\text{Horizontal run}} \\ &= \frac{3}{6} \\ &= \frac{1}{2} \end{aligned}$$



We usually leave the answer as a fraction.

The gradient is **positive** because the line **slopes up** to the right.

EXAMPLE

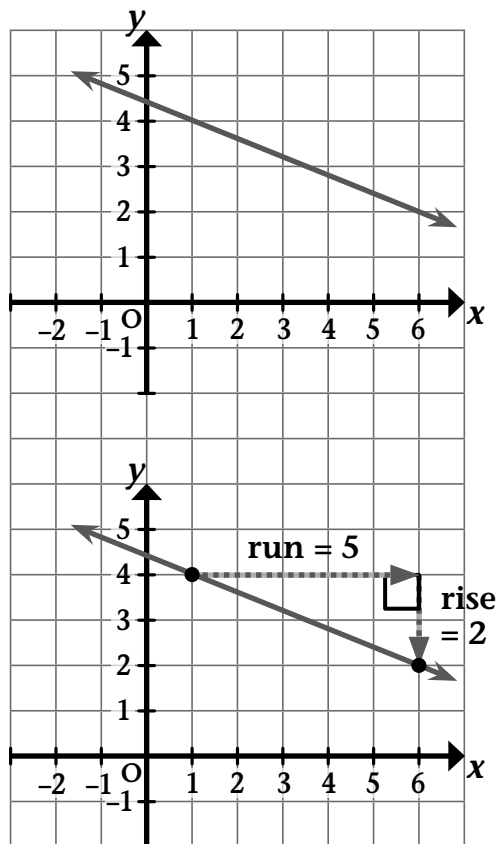
Calculate the gradient of this line.

Steps to follow

- Pick two points on the grid.
- Using grid lines, draw the right-angled triangle joining these two points.
- Count the rise and the run.
- Calculate the gradient.

$$\begin{aligned} \text{Gradient} &= \frac{\text{Vertical rise}}{\text{Horizontal run}} \\ &= \frac{-2}{5} \end{aligned}$$

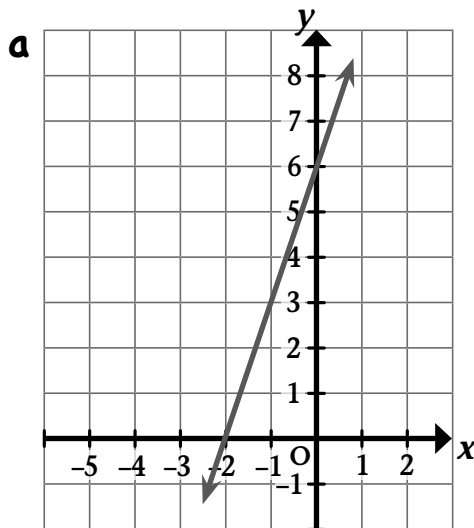
The gradient is **negative** because the line **slopes down** to the right.



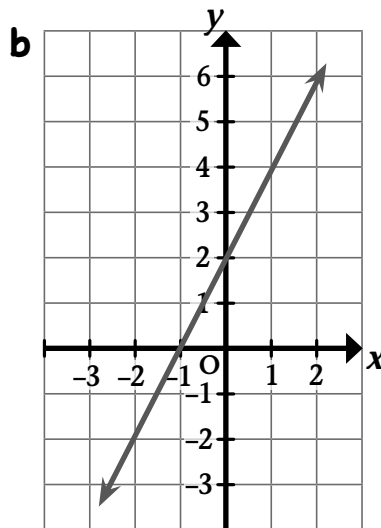
6.6 The gradient of a line B

CONTINUED

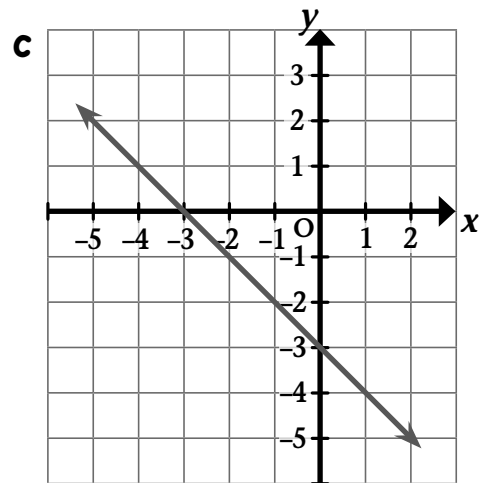
- 1 Calculate the gradient of each line.
Give your answer as a fraction in its simplest form.



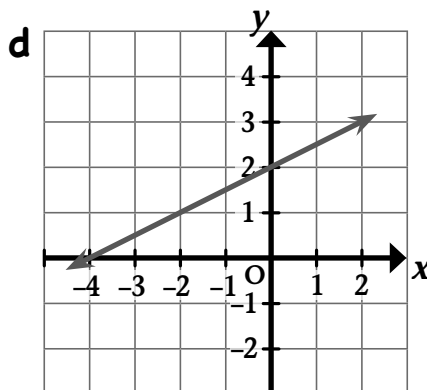
Gradient = _____



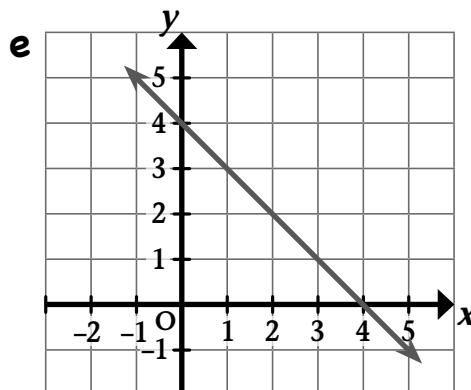
Gradient = _____



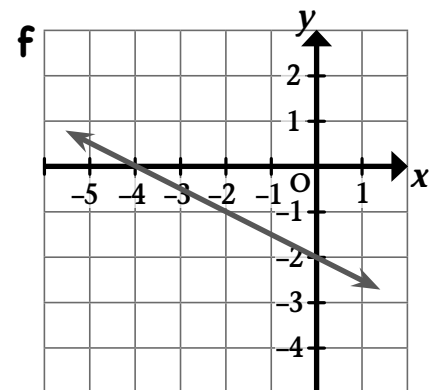
Gradient = _____



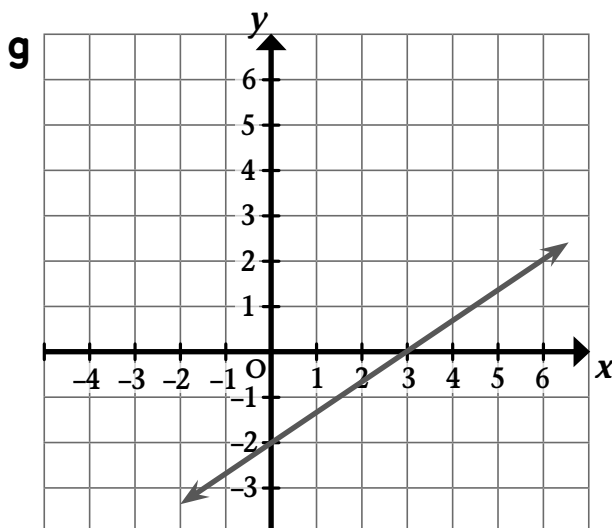
Gradient = _____



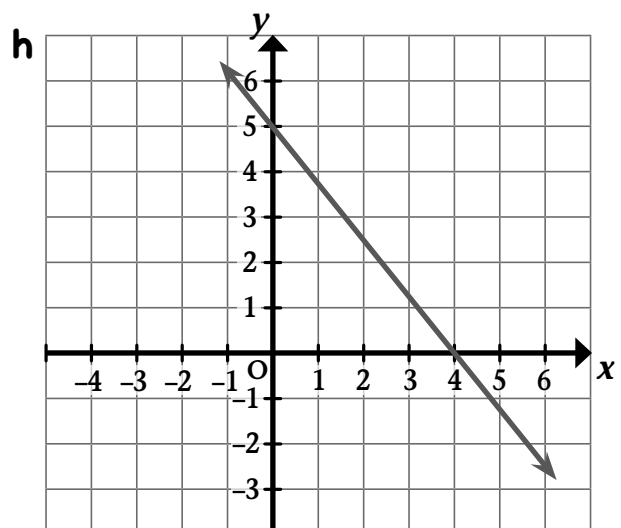
Gradient = _____



Gradient = _____



Gradient = _____



Gradient = _____

6.7 Calculating the gradient

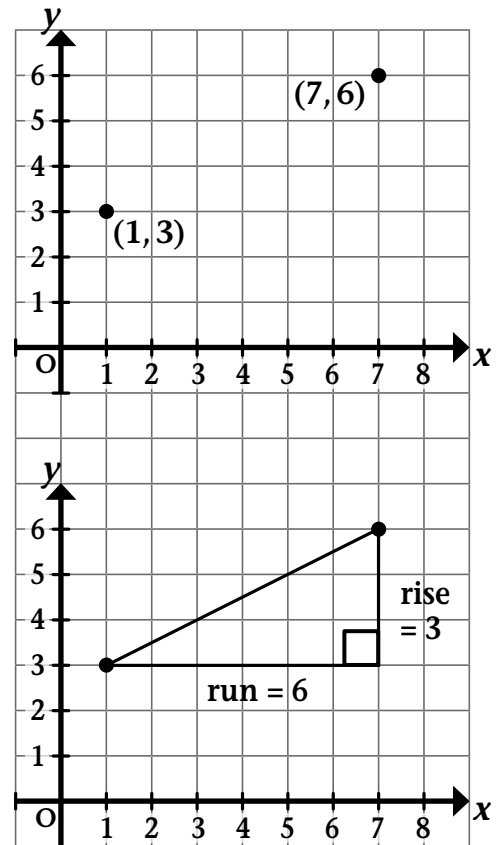
Rasheed wanted to calculate the gradient of the line joining the points (1, 3) and (7, 6).

He followed these steps:

- Plot the two points on a number plane.
- Join the two points with a straight line.
- Using grid lines, draw the right-angled triangle joining the two points.
- Count the rise and the run.
- Calculate the gradient using:

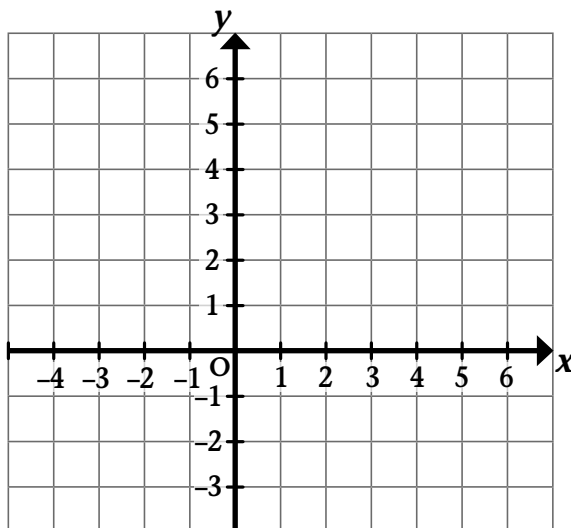
$$\text{Gradient} = \frac{\text{vertical rise}}{\text{horizontal run}}$$

$$\begin{aligned}\text{Gradient} &= \frac{\text{rise}}{\text{run}} = \frac{3}{6} \\ &= \frac{1}{2}\end{aligned}$$



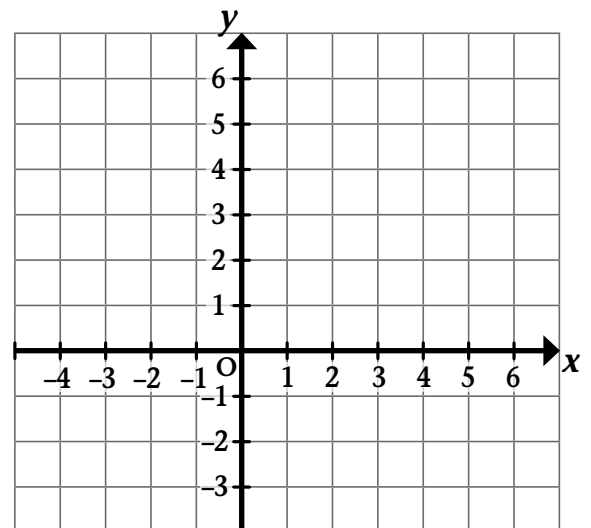
- 1** Plot each pair of points.
Calculate the gradient of the line joining them.

a (1, 2) and (4, 5)



Gradient = _____

b (-1, 0) and (5, 4)

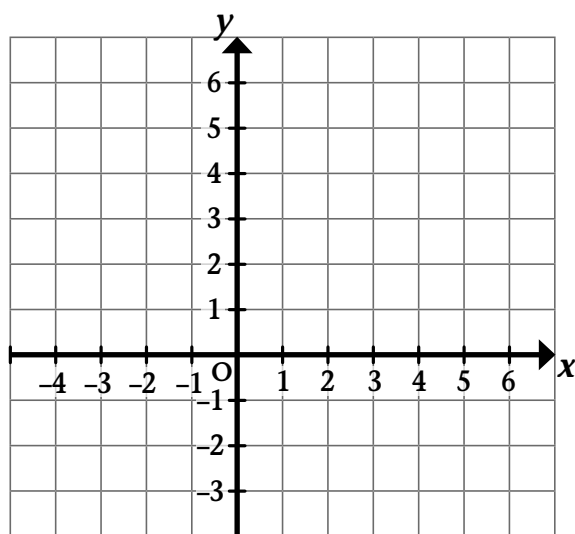


Gradient = _____

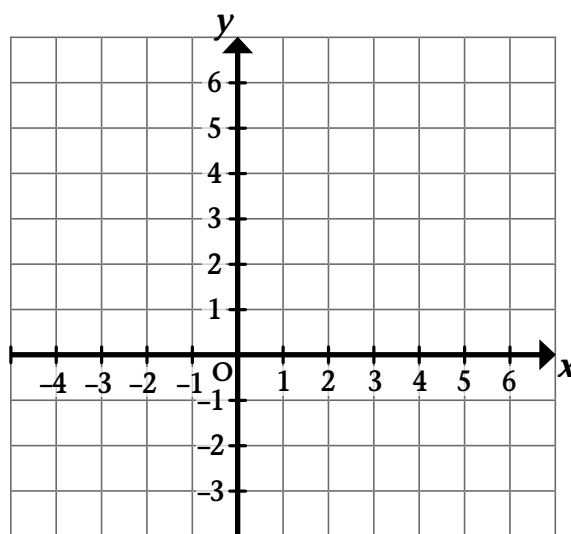
6.7 Calculating the gradient

CONTINUED

c (2, 5) and (5, 2)



d (-3, 3) and (5, -1)



Gradient = _____

Gradient = _____

Rasheed noticed that the **horizontal run** is equal to the **difference of the x co-ordinates**. The x co-ordinate is the first number in an ordered pair.

The **vertical rise** is equal to the **difference of the y co-ordinates** (the second numbers).

He wrote: $\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{\text{difference of the } y \text{ co-ordinates}}{\text{difference of the } x \text{ co-ordinates}}$

EXAMPLE

Calculate the gradient of the line joining (1, 3) and (7, 6).

$$\begin{aligned} \text{Gradient} &= \frac{\text{difference of the } y \text{ co-ordinates}}{\text{difference of the } x \text{ co-ordinates}} = \frac{6-3}{7-1} \\ &= \frac{3}{6} \text{ or } \frac{1}{2} \end{aligned}$$

2 Calculate the gradient of the line joining each pair of points.

a (4, 2) and (8, 5) gradient = _____

b (2, 6) and (5, 3) gradient = _____

c (-1, 2) and (5, 4) gradient = _____

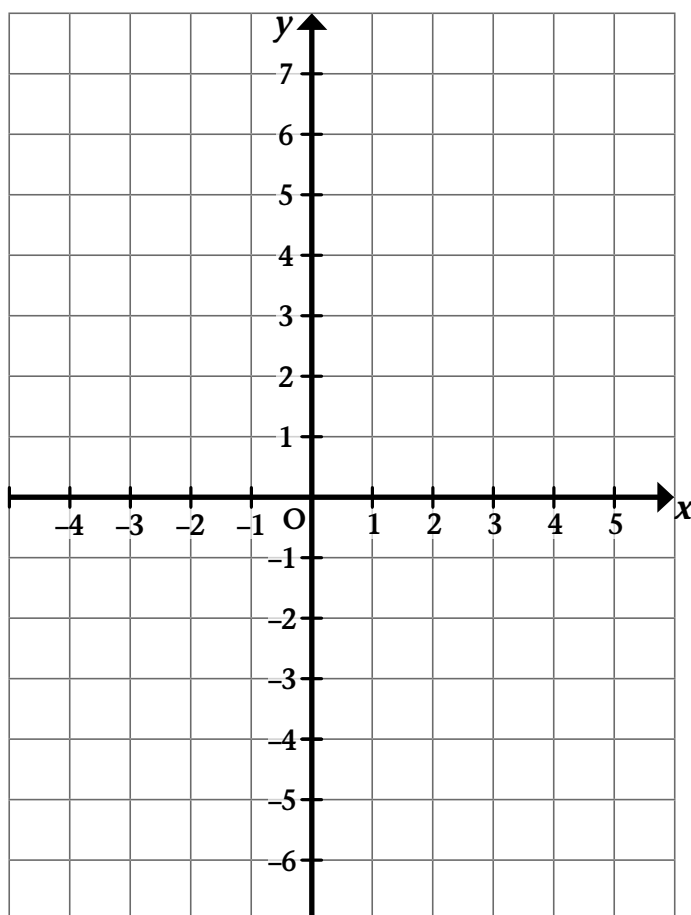
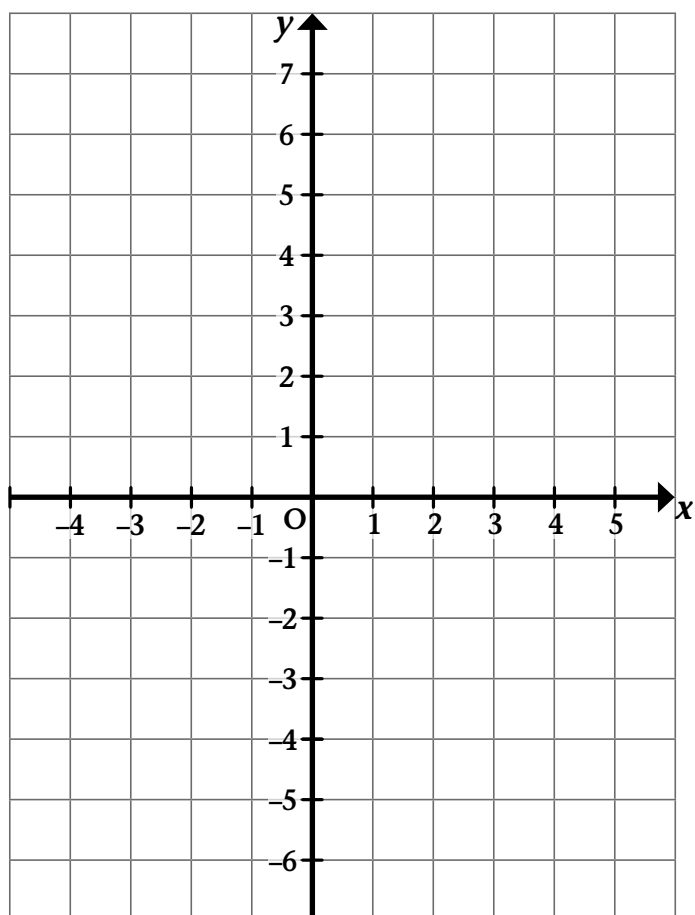
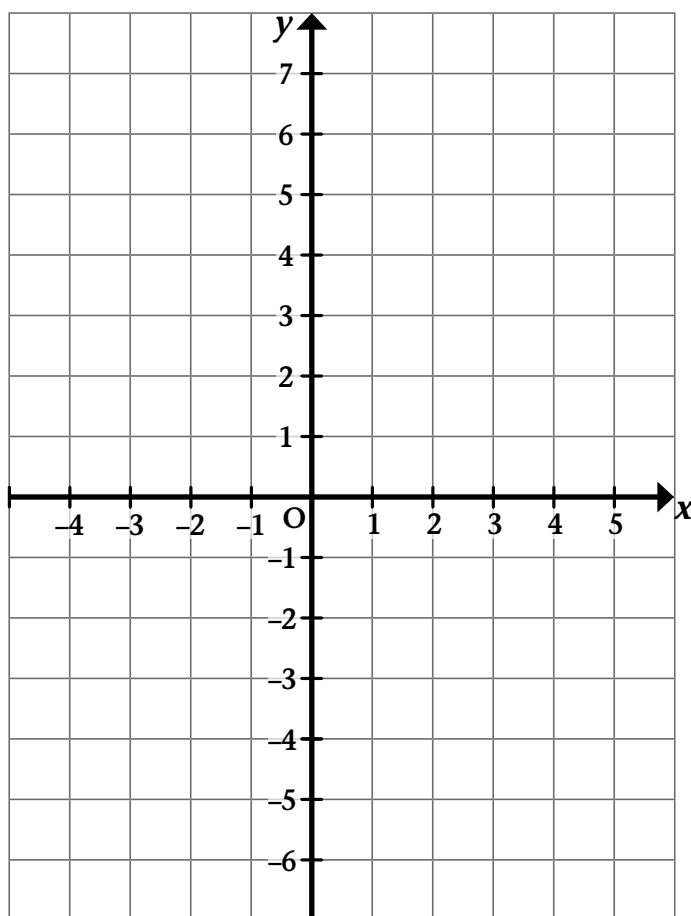
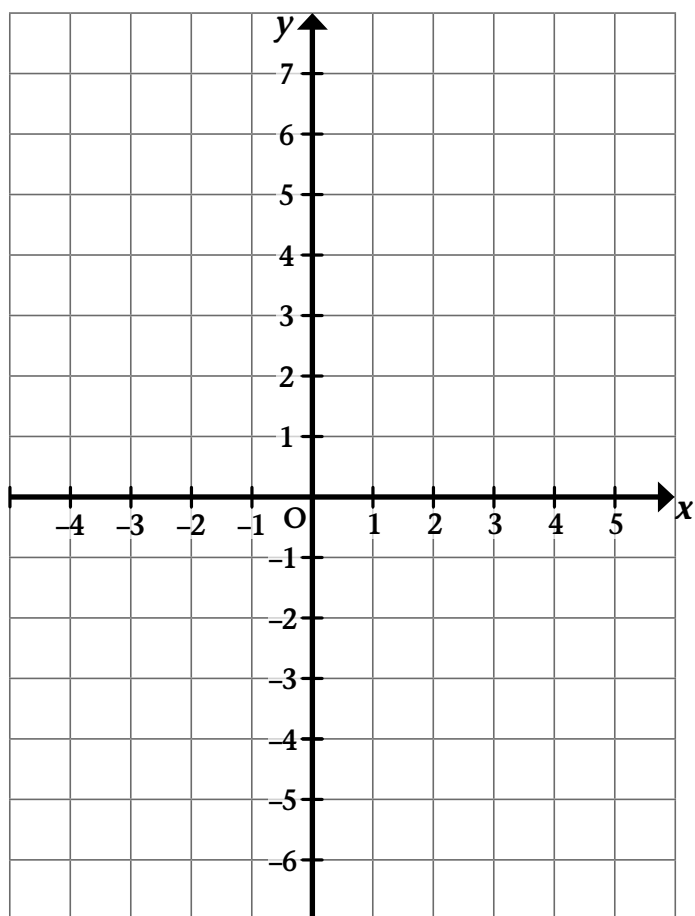
d (1, 1) and (-3, 6) gradient = _____

e (-4, -2) and (2, 1) gradient = _____

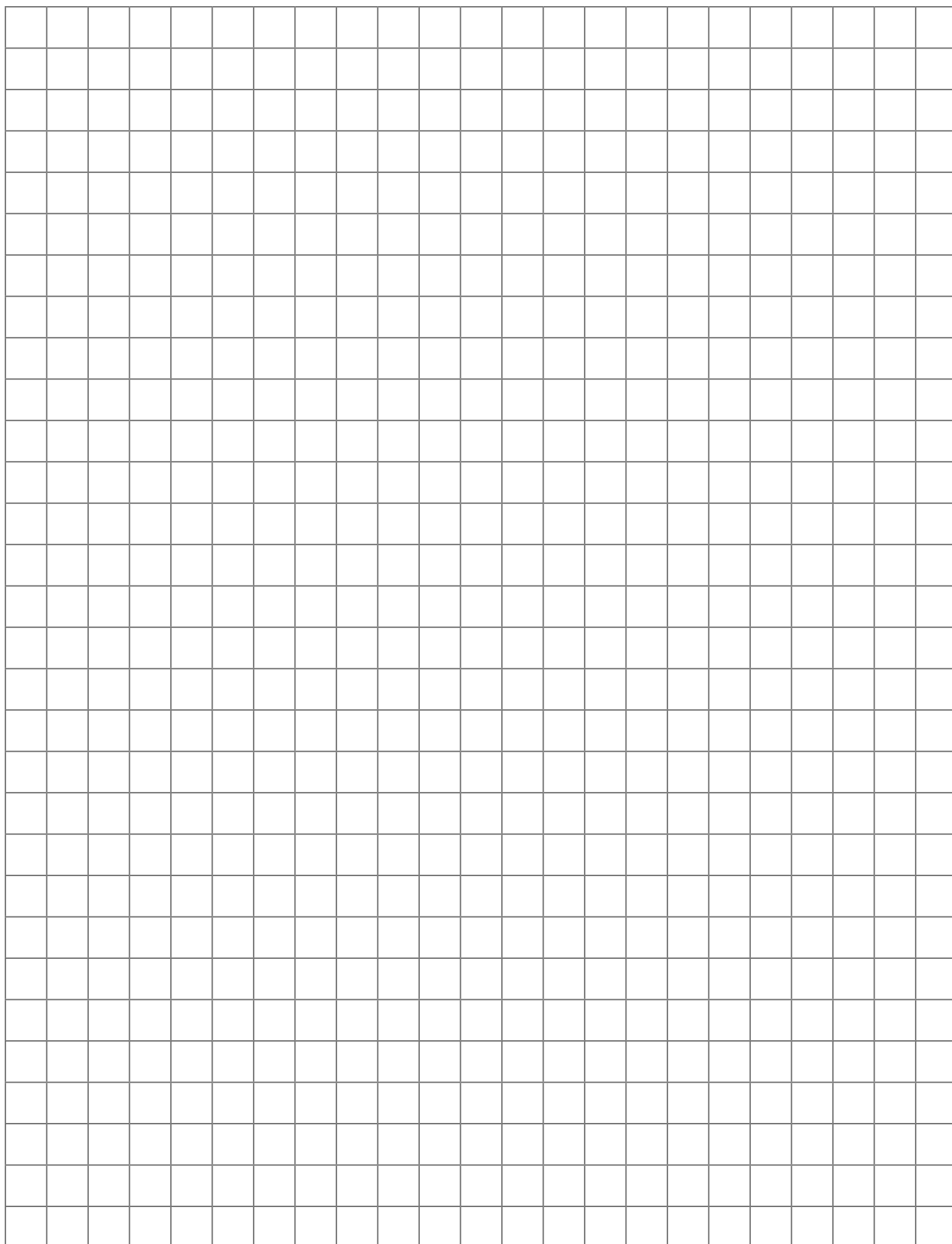
f (1, -1) and (5, -3) gradient = _____



Resource: number planes



Resource: 8 mm grid



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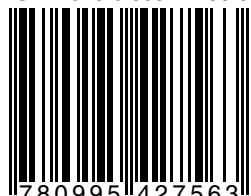
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